



Tribhuvan University
Asian School of Management and Technology

Final Year Project Report
On
“Leave Management System for Students”

Submitted to:

Office of Dean
Institute of Science and Technology

Under the Supervision of
Mr. Surya Bam

*In partial fulfillment of the requirement for the bachelor's degree in
Computer Science and
Information Technology*

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SUPERVISOR'S RECOMMENDATION

I hereby declared that this project is prepared under my supervision by Bishal Adhikari (29093/078), Sudip Khanal (29123/078) and Sujan Chaudhary (29125/078) entitled “Leave Management System” in partial fulfillment of the requirements for the degree of Bachelor in Computer Science and Information Technology be processed for the evaluation.

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CERTIFICATE OF APPROVAL

This project entitled “Leave Management System using Priority Algorithm” has approved and recommended to the department of Computer Science for acceptance and presented in a satisfactory manner to narrate its acceptance submitted by Bishal Adhikari, Sudip Khanal and Sujana Chaudhary as prerequisite to partial fulfillment for the degree of Bachelor of Science in Computer Science & Information Technology.

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With respect,

Bishal Adhikari

Sudip Khanal

Sujan Chaudhary

ABSTRACT

The purpose of the Leave Management System is to automate the existing manual system. By the help of computerized equipment and full-fledged computer software, fulfilling their requirements, so that their leave data/information can be stored for a longer period with ease of access. The required software and hardware are easily available and easy to work with. The Leave Management System, as described above, can lead to easy applying for leave and approval. It is a secure, reliable and fast management system. It can assist the user to concentrate on their other activities rather than facing the awkward situation of asking for leave manually. Thus it will help organizations in better utilization of resources. The organization can maintain computerized records without redundant entries. The aim is to automate its existing manual system by the help of computerized equipment and full- fledged computer software, fulfilling their requirements, so that their valuable data/ information can be stored for a longer period with easy accessing and manipulation of the same. Basically the project described how to manage for good performance and better services for the employees.

Keywords: Leave Management System, redundant, computerized equipment, full-fledged computer software.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
HTTPS	Hypertext Transfer Protocol Secure
ID	Identification
IO	Input Output
OCR	Optical Character Recognition
RFID	Radio-Frequency Identification
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

CHAPTER ONE:

INTRODUCTION

1.1 Introduction

The Leave Management System is a comprehensive web-based application designed to help educational institutions efficiently track and manage student leave requests and absences. It provides a structured and user-friendly platform where students can apply for leave digitally, while teachers and administrators can easily review, verify, and act on these requests in a streamlined manner. In the current academic environment where students are engaged in multiple assignments, classes, and activities having an organized and automated leave management process has become essential.

To ensure accuracy and discipline, the system now includes key validation features such as restricting the student's phone number to exactly 10 digits during registration, preventing incomplete or invalid entries. It also introduces predefined maximum leave durations for each leave type, ensuring that students submit requests within the allowed limits set by institutional policy. If a student applies for a leave exceeding the permitted duration, the system automatically notifies the higher authority, prompting direct communication between the student and senior administrators for further approval.

The system incorporates a priority-based scheduling algorithm to maintain fairness and efficiency in processing leave requests. Students can specify their leave reasons and durations, while teachers or supervisors review and approve them based on urgency, duration, and regulations. The system ensures data accuracy, security, and transparency, greatly minimizing errors or manual manipulation.

Additionally, the system evaluates students based on total working days, attendance history, and number of leaves taken. This helps institutions categorize students according to attendance performance, discipline, and eligibility for various academic or extracurricular activities. As a web-based platform, the Leave Management System is accessible to students, teachers, and administrators from anywhere, promoting convenience and reliable record-keeping.

In conclusion, the Leave Management System offers an efficient and transparent solution for automating leave processes within educational institutions. By simplifying leave applications, enforcing validation rules and providing administrators with effective monitoring tools, the system enhances accuracy, accountability and overall institutional productivity.

1.2 Problem Statement

While studying in an institution, it becomes really difficult for the administration to track students' leaves and attendance. It is also a problem and an awkward situation for a student to approach their teacher or class coordinator every time they need to request leave.

1.3 Objectives

- To simplify the process of requesting and approving student leaves.
- To track and manage student leave records accurately and efficiently.
- To reduce errors and delays caused by manual leave management.

1.4 Scope and Limitation

The scope of leave management system are as follows:

- Automating leave application and approval process.
- Streamlining record keeping of student leave.
- Maintaining accurate and updated leave balance.
- Providing easy access to leave history and status.

Every system has error so this system has also limitations are as follows:

- Limited flexibility in accommodating non-standard leave requests.
- Potential for errors in data entry and processing.
- Inability to address complex leave scenarios that require human intervention.
- Dependence on system availability may not work properly without internet or power.
- Requires user training for effective usage and understanding.

1.5 Development Methodology

Agile Methodology was used as a software development process framework that utilizes an iterative approach, open collaboration and process adaptability throughout the project's life cycle, reducing overall risk and allowing the project to adapt to changes more quickly. The main steps followed are:

- **Defining Key Features:** Start by outlining the most important functions such as user login, leave application, leave approval, and leave balance tracking, ensuring they are clearly defined and practical.
- **Building the First Version:** Develop the basic modules like user registration, authentication, leave request submission, and admin approval interface. Test these features and share the initial version for early feedback.
- **Review and Plan Ahead:** Collect feedback from the supervisor and team, identify what needs to be improved, and prepare for the next sprint with updated priorities.
- **Adding More Features in Stages:** Gradually introduce advanced parts of the system such as leave history reports, role-based dashboards, and attendance integration, while continuously testing for bugs or errors.
- **Ongoing Updates and Support:** Continue improving the system by fixing issues, enhancing security, refining the user interface, and ensuring the application remains reliable, efficient, and user-friendly.

1.6 Report Organization

The Report Organization describes how the study is structured and conducted. This report is divided into six chapters, each addressing different aspects of the study. Chapter One presents the introduction, statement of the problem, objectives of the study, scope and limitations, development methodology and organization of the study. Chapter Two focuses on the background study and review of related literature, providing the necessary theoretical foundation. Chapter Three discusses system analysis, including requirement analysis, feasibility study, data modeling, and process modeling of the system. Chapter Four deals with the design of the system, particularly the system architecture and its components. Chapter Five explains the implementation phase, the algorithms used, and the tools applied, along with the testing of the system both as individual units and as a

complete system. Finally, Chapter Six provides the conclusions drawn from the study, highlights its limitations and offers recommendations for future improvements.

CHAPTER TWO:

BACKGROUND STUDY AND LITERATURE REVIEW

2.1. Background Study

The need for a Leave Management System arises from the limitations of traditional manual methods such as paper forms, emails, and spreadsheets. These older approaches often cause delays, errors, missing records, and difficulties in tracking student attendance and leave patterns. As institutions grow, managing leave manually becomes inefficient and unreliable.

A digital Leave Management System addresses these issues by automating the entire process. Students can submit leave requests online, and the system ensures accuracy through features like 10-digit phone number validation during registration and fixed leave duration limits based on leave type. If a student applies for leave beyond the allowed duration, the system automatically alerts higher authorities, reducing misuse and ensuring proper oversight.

Teachers and supervisors can review and act on requests easily, while the system maintains real-time records, improves transparency, and reduces administrative workload. Overall, this system enhances efficiency, data accuracy, and accountability in managing student attendance and leave.

2.2. Literature Review

The Leave Management System is a web-based application designed to automate the process of managing and tracking different types of student leave requests and absences. It enables students to easily submit leave requests, cancel previously made applications, check the current status of their submissions, and view detailed records of approved or rejected leaves [1]. The system maintains a centralized database that stores each student's leave history, attendance records, and related documentation, ensuring that all data remains accurate, up-to-date, and easily retrievable when needed [2]. This automation minimizes human error, reduces administrative workload, and speeds up decision-

making, providing a more transparent and efficient leave-handling mechanism for educational institutions [3].

Traditionally, teachers and administrative staff maintained student leave records manually in registers, which was inefficient, time-consuming, and prone to loss, duplication, and record tampering [4]. With advancements in information and communication technology, automated systems have replaced manual processes, improving reliability, data integrity, and accuracy in tracking attendance and leave activities [5]. Similar to professional tools such as Calamari and actiPLANS, widely used in organizations to manage employee leave requests, schedules, and approvals [6], this proposed system adapts such functionalities for the academic environment. It allows students to interact with the system through a secure login, submit applications online, upload supporting documents, and receive real-time notifications once a decision is made. Teachers or supervisors, in turn, can easily view, approve, or decline requests, making the overall process faster, more structured, and more organized [7].

The system incorporates priority preemptive scheduling to ensure fairness and timely decision-making when multiple leave requests are pending simultaneously. It also evaluates student attendance patterns based on the number of attended working days, frequency of leave submissions, and document validity. Through this analysis, the system categorizes students according to their attendance behavior, punctuality, level of discipline, and eligibility for academic activities such as exams, events, and practical assessments [8]. Additionally, the system generates analytical reports that provide insights into attendance trends, peak leave seasons, and departmental leave patterns, aiding administrators in better planning and resource management.

In conclusion, the Leave Management System serves as a secure, scalable and user-friendly platform that digitizes traditional leave management processes and brings efficiency to institutional operations. It not only enhances accountability, transparency and fairness in handling student leave records but also provides accurate data analytics for better academic decision-making [9].

CHAPTER THREE:

SYSTEM ANALYSIS

3.1 System Analysis

System analysis involves studying the system in detail to understand how it works and what improvements are needed. It focuses on identifying problems, gathering data, and separating the system into smaller parts to make it easier to study. By analyzing each part and how they connect, we can clearly see what the system is expected to do. In this way, system analysis acts as a method of problem solving, helping to check whether each component works properly and contributes toward the overall goal of the system.

3.1.1 Requirement Analysis

To develop the Leave Management System, we identified both functional and non-functional requirements that the system must fulfill.

3.1.1.1 Functional Requirements

Functional requirements are the features and functions that the Leave Management System must provide.

- **User Registration and Authentication:** Only registered students can access the system. Admins have extended privileges to monitor system activities and manage teacher assignments.
- **Leave Application Form:** Students can submit leave requests online, providing details such as leave type, start and end dates, reason for leave, and uploading supporting documents.
- **Priority Scheduling:** The system analyzes submitted leave requests and applies the Priority Scheduling algorithm to determine the order in which requests are forwarded to teachers.
- **Teacher Approval:** Teachers review leave requests and supporting documents, then approve or reject requests based on the details and urgency.
- **Status Update Module:** The system updates leave status in the database and notifies students of approval or rejection, providing reasons in case of rejection.

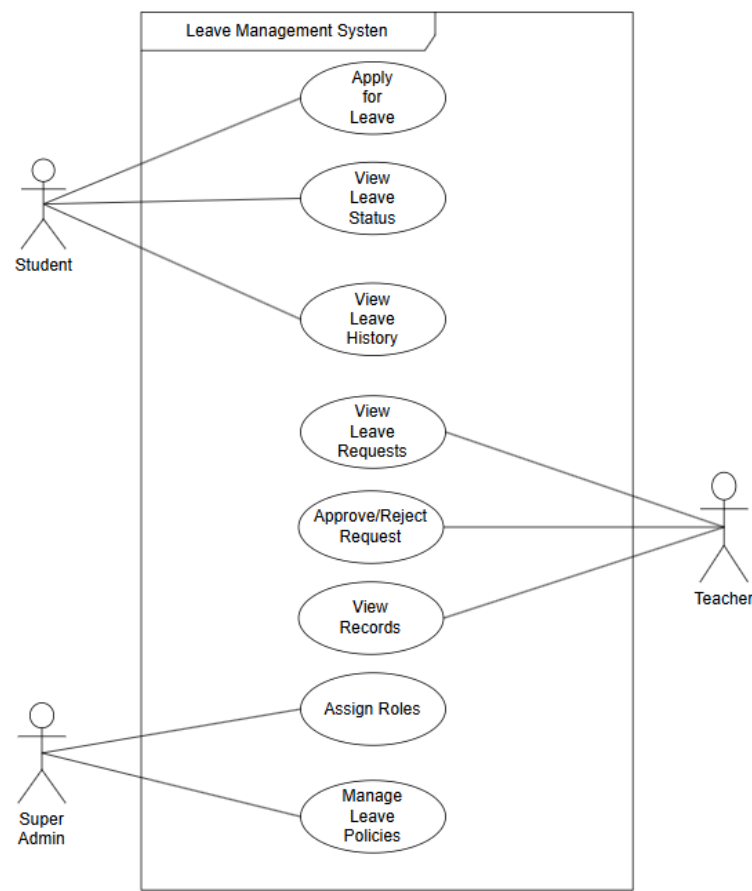


Figure 3.1: Use Case Diagram of Leave Management System

3.1.1.2 Non-Functional Requirement

Non-functional requirements are the requirements that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors. They are contrasted with functional requirements that define specific behavior or functions. The non-functional requirements of the system are:

- **Usability:** The interface should be simple, clean and easy to use for all users (admin, teacher and student).
- **Reliability:** System is accurate, as we don't use many components and thus provides the user specified results in specific time duration.
- **Performance:** System should handle multiple user requests simultaneously without slowing down.

- **Scalability:** The system should be designed so it can accommodate more users or institutions in the future without major redesign.
- **Security:** All sensitive data must be securely encrypted during storage and transmission. Only authorized users can access certain features, and all communication must be encrypted using HTTPS.
- **Maintainability:** The system should be easy to update and maintain, with a clear and modular architecture that supports future enhancements and feature upgrades.

3.1.1.3 Dataset Description

For this project, a custom synthetic dataset was created to test and validate the Leave Management System workflow, particularly for leave request prioritization and processing. The dataset contains 500 records with 8 attributes, designed to simulate student leave applications in an academic environment.

The attributes include both categorical and numerical features describing each student's profile, leave details and document submission status. The attributes are as follows:

- **Student ID:** Unique identifier for each student.
- **Gender:** Indicates the student's gender (Male or Female).
- **Department:** Academic department of the student (e.g., Science, Management, Engineering).
- **Leave Type:** Type of leave requested (Medical, Personal, Emergency, Other).
- **Leave Duration:** Number of days requested.
- **Submission Date:** Date when the leave application was submitted.
- **Document Status:** Indicates whether supporting documents were provided and valid (Valid/Invalid).
- **Leave Status:** Target variable showing whether the leave was Approved or Rejected.

The Leave Status attribute serves as the dependent variable, while the remaining attributes act as independent variables. Categorical values were encoded numerically to make them compatible with system testing and prioritization algorithms.

This dataset was manually prepared to simulate realistic student leave applications, providing a practical basis for testing system workflow, document verification, and the Priority Scheduling algorithm.

3.1.2 Feasibility Analysis

The proposed project's viability is determined by the feasibility study:

i) Technical Feasibility

This is web-based application which uses priority based scheduling to schedule list of employee's leave. The leave can be approved on the basis of leave's priority or the number of leave days that employee has requested for. Since the system is simple and don't require any external devices for functioning we will be able to construct this system using resources available in internet and our existing knowledge of technologies.

ii) Operational Feasibility

This system aims at making it easier for students to request time off and for managers to approve or deny those requests. For that the system must be accessible to student and supervisor working at the particular institution. The institution must implement the system. After implementation the system can be easily used. Apart from this, no prerequisite is required.

iii) Economic Feasibility

This system will be constructed at a minimum price. Almost every resource required for the system will be acquired from the internet. We will be able to do this project using our understanding of available languages and technologies.

iv) Schedule Feasibility

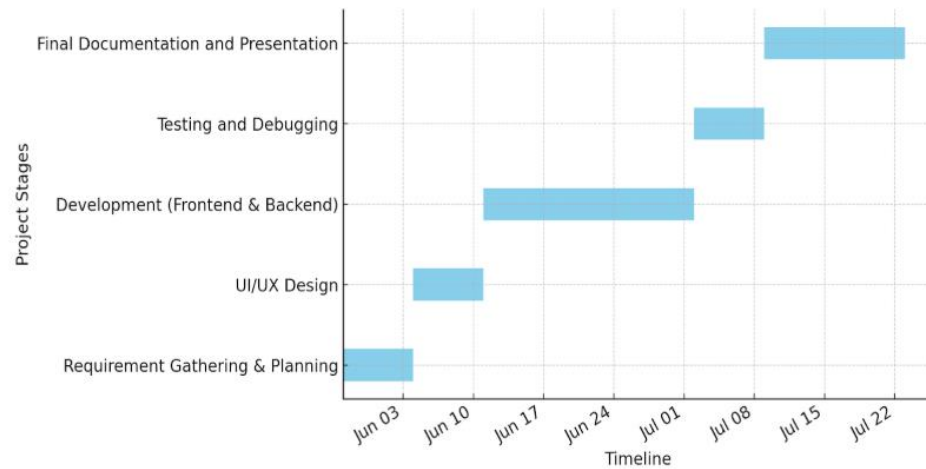


Figure 3.2: Gantt Chart

The Gantt chart illustrates the timeline of the Online Leave Management System project. The development began with Requirement Gathering and Planning from June 3 to June 9, 2025, where objectives and system scope were defined. This was followed by the UI/UX Design phase from June 10 to June 15, 2025, focusing on creating a simple and user-friendly interface. The Development (Frontend & Backend) phase ran from June 16 to July 3, 2025, during which the main system modules for leave request, approval, and tracking were implemented. Next, Testing and Debugging took place from July 6 to July 12, 2025, ensuring the system's reliability and performance. Finally, the Final Documentation and Presentation phase from July 15 to July 25, 2025 summarized the entire development process and outcomes.

3.1.3 Analysis

3.1.3.1 Object Modeling Using Class Diagram

The class diagram models the main objects in the Leave Management System and how they relate to one another.

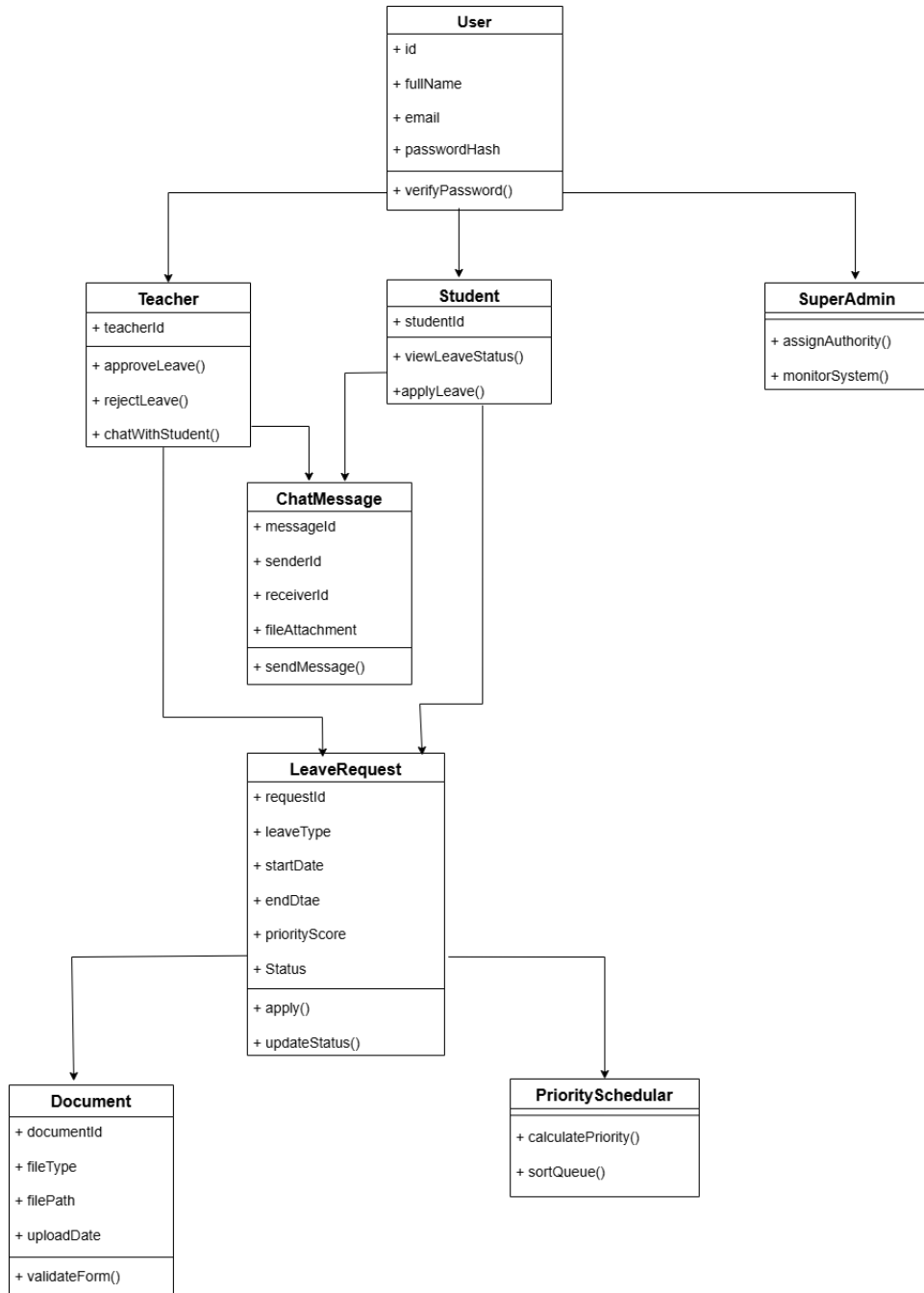


Figure 3.3: Class Diagram of Leave Management System

The class diagram consists of eight main classes. The central base class is **User**, which stores core account information such as `id`, `fullName`, `email`, and `passwordHash`, and provides the method `verifyPassword()` to authenticate users securely. Three specialized roles extend this class: **Student**, who can apply for leave and view the status of previous requests; **Teacher**, who can review submitted requests, approve or reject them and

communicate with students and SuperAdmin, responsible for assigning authority and monitoring the overall system.

The primary business entity is LeaveRequest, which manages leave details including requestId, leaveType, reason, startDate, endDate, priorityScore, and status. Each request may include one or more uploaded Document objects. A Document contains attributes like documentId, fileType, filePath, and the uploadDate, while validateForm() ensures that students submit valid and properly formatted files, such as medical certificates.

Communication between students and teachers is handled by the ChatMessage class, which stores the messageId, senderId, receiverId, timestamp, and any fileAttachment for sharing evidence or supporting documents. To make the approval process fair and efficient, the system uses a PriorityScheduler, which calculates priority values and sorts pending leave requests based on urgency, reason, or leave type.

Overall, this structure supports secure authentication, digital document handling, real-time communication, and a prioritized approval workflow across students, teachers, and administrators within the institution.

3.1.3.2 Dynamic modeling using Sequence Diagram

Sequence diagram in the UML is a type diagram that illustrates the sequence of messages between objects in an interaction. It consists of a group of objects represented by lifelines and messages they exchange during the interaction denoted by arrow symbols.

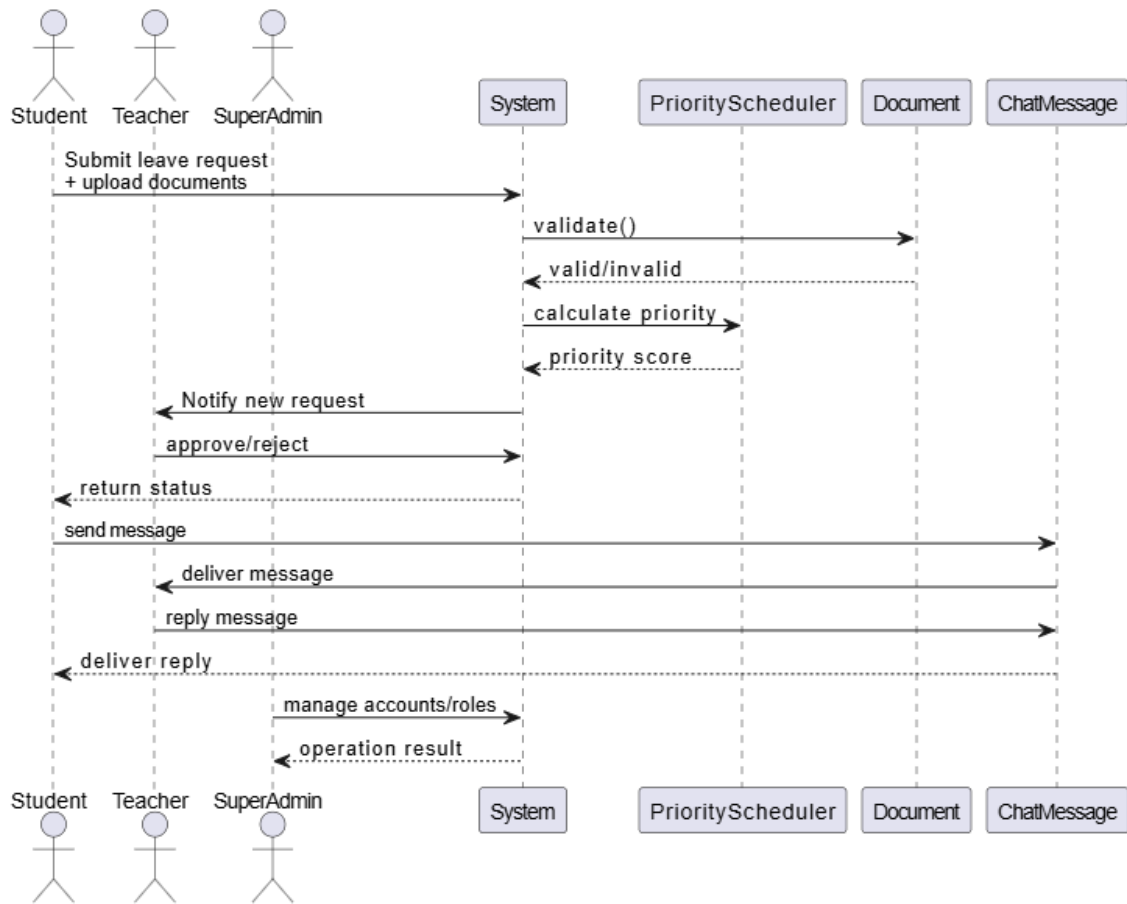


Fig 3.4: Sequence Diagram of Leave Management System

Figure 3.4 represents the sequence diagram of the Leave Management System. It includes the Student, System, Document module, Priority Scheduler, Teacher, ChatMessage service and SuperAdmin.

The process starts when the Student submits a leave request with supporting documents. The System receives the request, stores the details, and sends the uploaded files to the Document module for format and validity checks.

- If the document is invalid, the request may be rejected and the Student is notified.
- If valid, the System sends the request to the Priority Scheduler.

The Priority Scheduler assigns a priority score based on criteria such as urgency and leave type, then returns it to the System. After this, the Teacher is notified and reviews the request, deciding to approve or reject it. The System updates the status and sends the result back to the Student.

If clarification is needed, the Student and Teacher can communicate through the ChatMessage service, which handles message delivery and attachments. Meanwhile, the SuperAdmin can perform administrative tasks such as managing roles, and the System returns the operation results.

Overall, the diagram shows a structured workflow: verify documents, assign priority, review by teacher, communicate when needed and allow administrative control by the SuperAdmin.

3.1.3.3 Process Modeling using Activity Diagram

The activity diagram in UML shows the flow of actions in a system and how control moves from one step to the next, with clear points for choice, repetition, or parallel work.

Figure 3.5 illustrates the workflow of the Leave Management System, showing how students, the system, teachers and the admin interact. The process begins with the system initializing, after which students can register or log in.

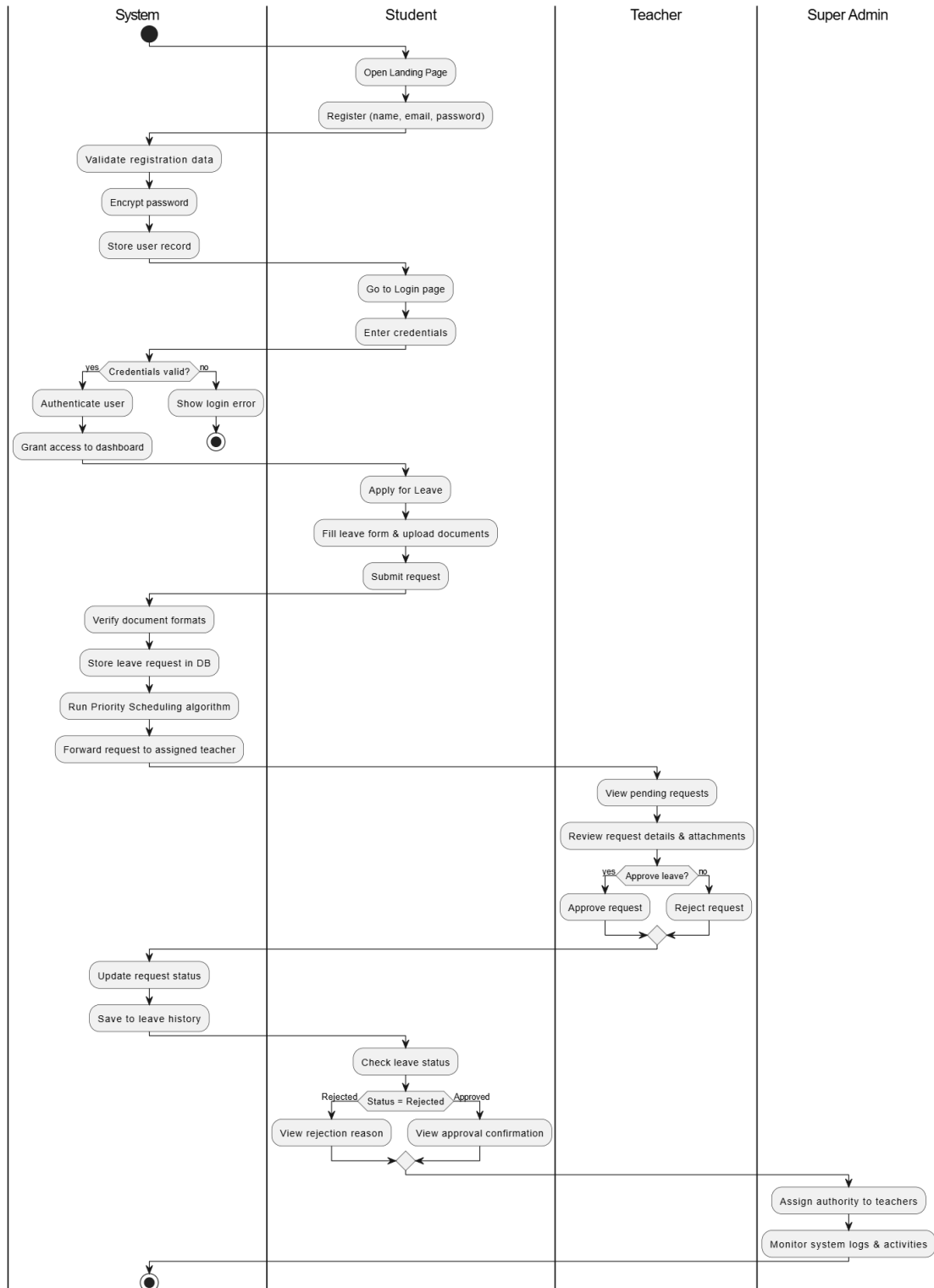


Figure 3.5: Activity Diagram of Leave Management System

The system validates credentials and grants access to the dashboard upon successful authentication. Students can then apply for leave by filling out the form and uploading documents. The system verifies the documents and applies a Priority Scheduling

algorithm to forward requests to the appropriate teacher. Teachers review requests and either approve or reject them. The system updates the leave status and records it in history. Students can check the outcome, while super admins monitor activities and assign authorities. The diagram ensures a clear, logical, and secure flow from registration to leave approval or rejection.

CHAPTER FOUR:

SYSTEM DESIGN

4.1 Design

System Design includes the means and methodologies to improve the management and control of the software development process. It includes structuring and simplifying the process using several diagrams and standardizing the development process by specifying the required activities and techniques to be implemented.

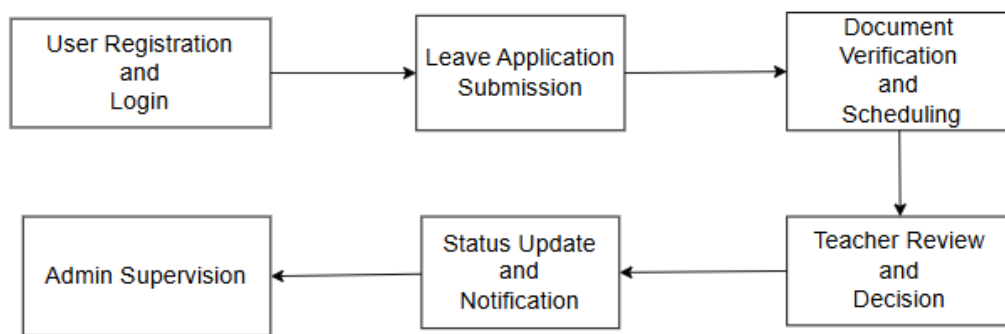


Fig 4.1: High-Level System Design of Leave Management System

The system architecture is shown in Figure 4.1. The procedures involved are:

User Registration and Login

The process begins when a student registers a new account or logs in with existing credentials. During registration, students provide details such as name, email, 10-digit phone number and password. The system validates all inputs, encrypts the password, and securely stores the user record. After a successful login, students access their dashboard to apply for leave and view request history.

Leave Application Submission

Logged-in students can submit leave requests through the application form. They must specify the leave type, start and end dates, reason for leave and any supporting documents (e.g., medical certificates). The system automatically checks whether the requested duration stays within the permitted limit for that particular leave type. If the leave exceeds

the allowed period, the system redirects the request to the higher authority and notifies them accordingly.

Document Verification and Scheduling

Uploaded documents are validated for correctness and clarity. The system then applies the Priority Scheduling algorithm to arrange leave requests based on urgency, leave type rules, and submission time. After verification and scheduling, requests move to teachers or higher authorities for review as per the leave duration policy.

Teacher / Higher Authority Review

Teachers (or higher authorities in special cases) review pending leave requests, examine the provided details and documents, and make an approval or rejection decision. Their response is securely recorded in the system, ensuring fairness and proper documentation.

Status Update and Notification

Once a decision is made, the system updates the leave status in the database. Students can view approved, pending, or rejected requests on their dashboard. The system automatically sends notifications, including reasons for rejection or information about escalations to higher authority.

Admin Supervision

Administrators oversee the entire workflow, manage teacher roles, monitor system activity, validate institutional policies (like leave duration limits), and ensure smooth operation. They can review logs, handle escalated requests, and maintain overall system integrity.

4.1.1 Refinement of Class Diagram

The class diagram shows the main parts of the Leave Management System and how they relate to each other.

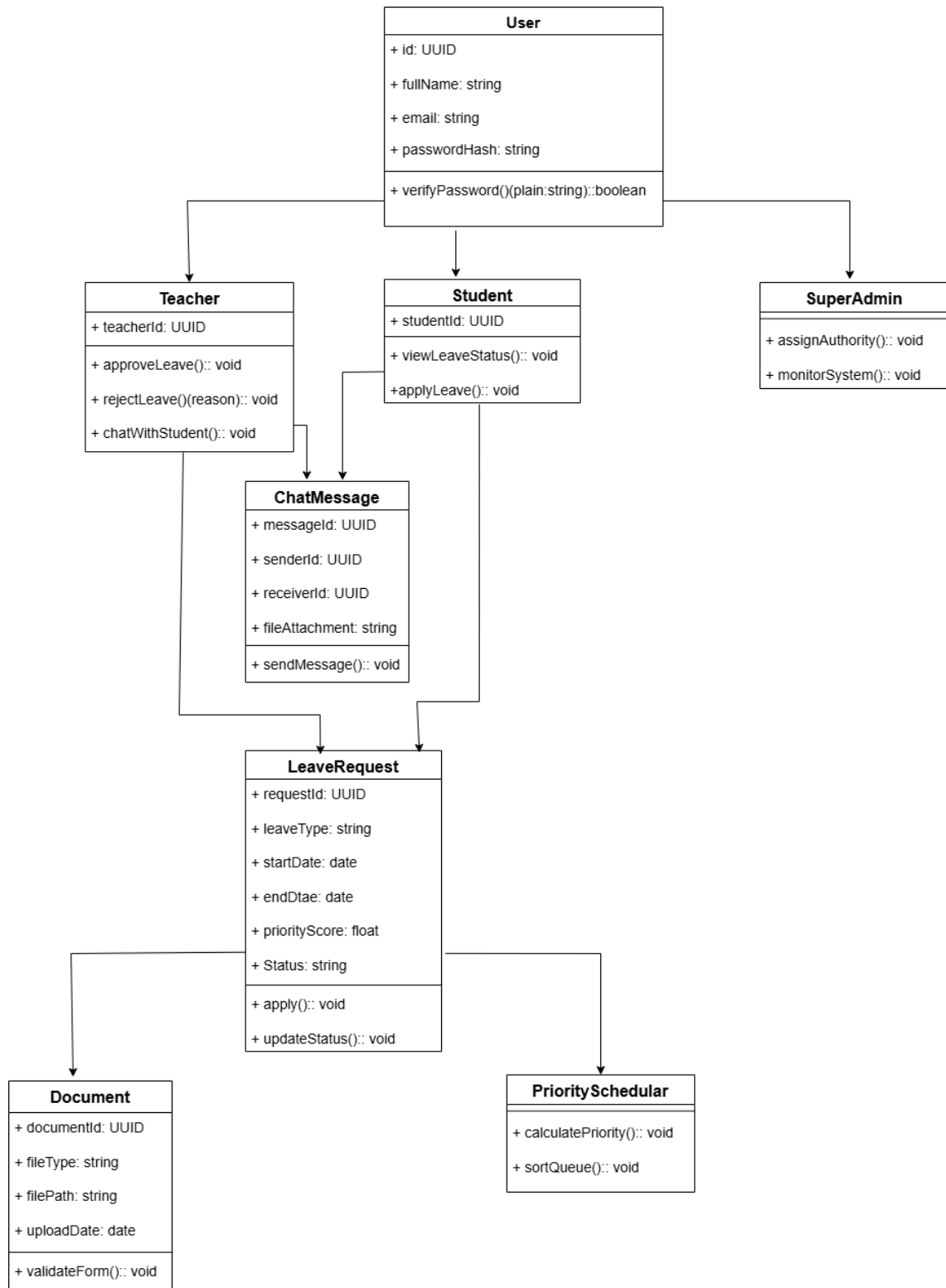


Fig 4.2: Refined Class Diagram of Loan Management & Approval System

At the center of the system is the User class, which holds basic account details such as ID, full name, email, and a hashed password. The verifyPassword() method compares an

entered password against the stored hash to authenticate the user. Three specialized types extend from it: Student, Teacher, and SuperAdmin.

The Teacher can review, approve or reject student leave requests and communicate through the internal chat system. The Student can submit leave applications and track their status. The SuperAdmin manages permissions and oversees system activity. Communication between users is handled by the ChatMessage class, which stores message details, optional attachments, and uses sendMessage() to deliver messages between users.

Leave submissions are represented by the LeaveRequest class. It stores leave type, date range, priority level, and current status. The methods apply() and updateStatus() guide the request through different stages like pending, approved, or rejected. Any supporting files are managed by the Document class, which stores document type, file path, and upload date. The method validateForm() checks if the document meets required standards. To maintain fairness, the PriorityScheduler class ranks leave requests. Its calculatePriority() and sortQueue() methods decide which requests should be addressed first based on urgency or criteria provided.

4.1.2 Refinement of Sequence Diagram

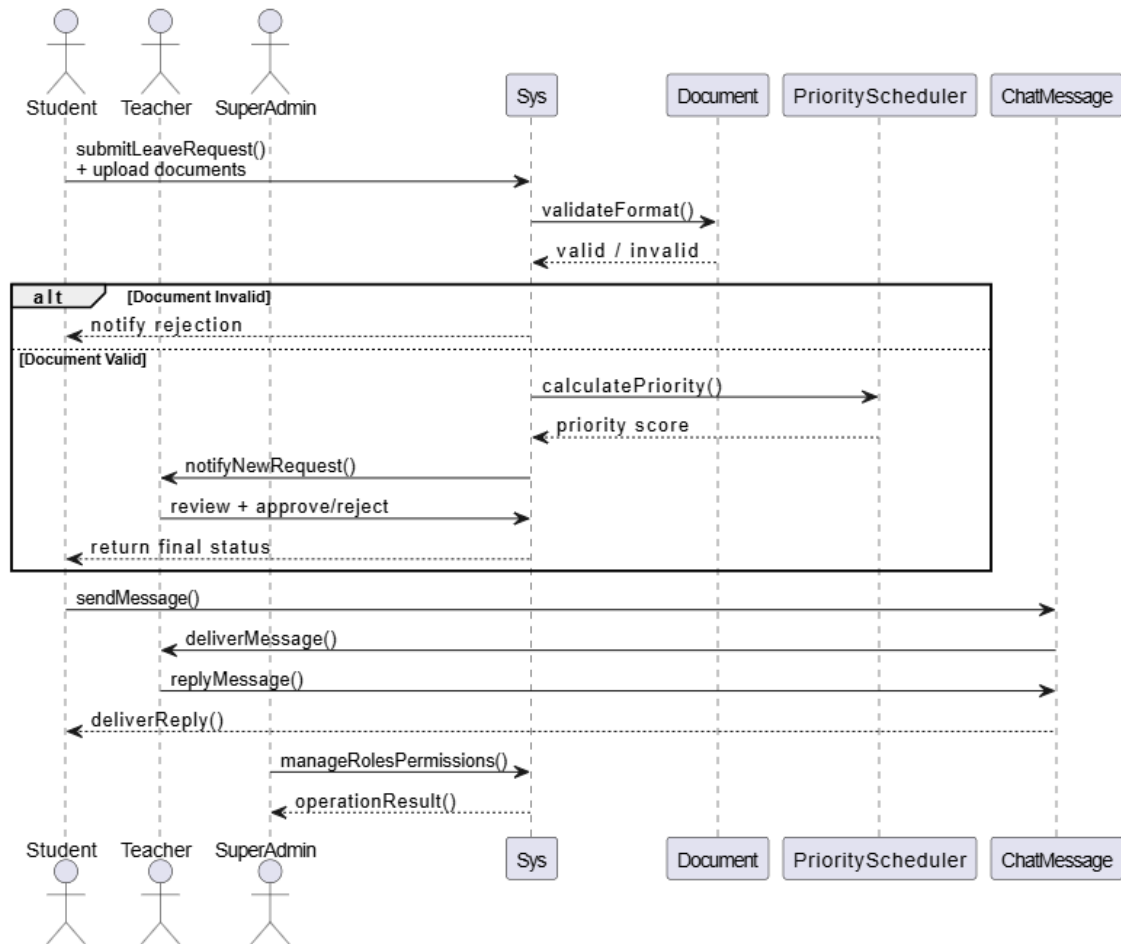


Fig 4.3: Refined Sequence Diagram of Loan Management & Approval System

The refined sequence diagram in Figure 4.2 presents a comprehensive view of the interactions between different system components during the entire leave management process. It illustrates each step, from the initial leave request submission by the student to the final status update, ensuring clear visibility of message flows between modules.

The process begins with the Student, who submits a leave request along with any required supporting documents. The System (Sys) first sends these uploaded documents to the Document module for format validation. If the documents are invalid, the system immediately notifies the student about the rejection, halting further processing. However, if the documents pass validation, the PriorityScheduler calculates the priority of the leave request based on predefined criteria, returning a priority score to the system.

Following this, the system notifies the Teacher of the new leave request. The teacher reviews the request and makes a decision to approve or reject it. The system then communicates the final decision back to the student, completing the leave request workflow. The diagram captures chat communication between the student and the teacher. When the student sends a message, it is handled by the ChatMessage module, which delivers it to the teacher. The teacher can reply, and the module ensures the response is delivered back to the student, facilitating smooth real-time communication.

Finally, the SuperAdmin role is included to manage system-wide roles and permissions. The SuperAdmin interacts with the system to perform administrative tasks, and the system returns the results of these operations, maintaining proper governance and access control. This refined sequence diagram demonstrates how the system ensures accuracy, structured communication and timely updates to all involved users, maintaining transparency and efficiency throughout the process.

4.1.3 Refinement of Activity Diagram

The refined activity diagram presents the detailed workflow of the Leave Management System, illustrating each step from user actions to the final outcome.

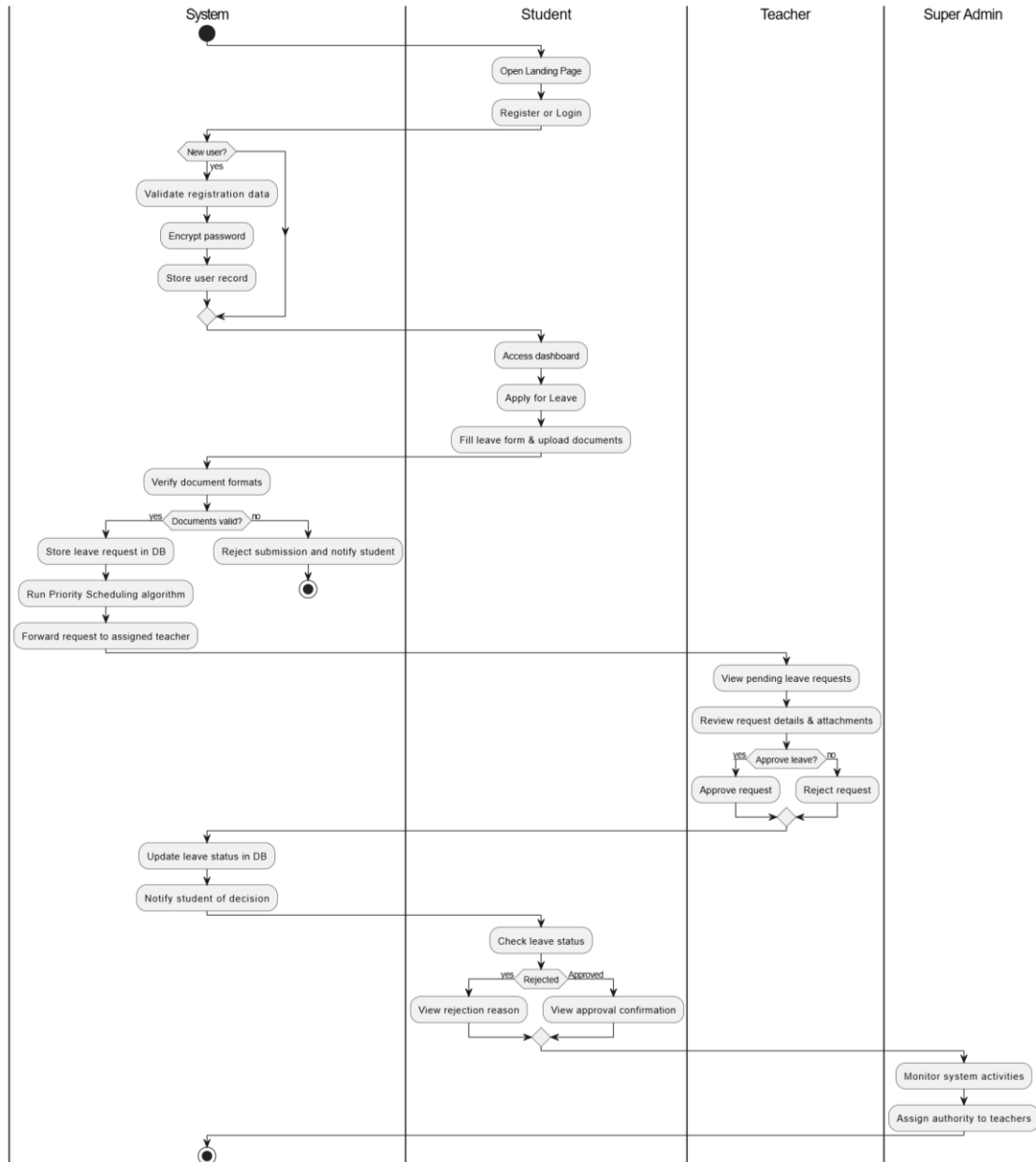


Fig 4.4: Refined Activity Diagram of Leave Management System

Figure 4.4 presents the refined activity diagram of the Leave Management System, showing how the Student, System, Teacher and Admin interact from registration to leave decision.

The process begins when a Student registers a new account or logs in. The system validates inputs, encrypts the password, and stores the user record securely. After authentication, the student accesses the dashboard and submits a leave request by completing the form and uploading supporting documents.

The System verifies the documents for correctness and authenticity. Invalid submissions prompt the student to correct or re-upload them, while valid requests proceed to the scheduling stage, where the Priority Scheduling algorithm determines the order in which requests are forwarded to the assigned teacher.

Teachers review pending requests and approve or reject them. The System updates the leave status, records it in the database, and notifies the student. Approved requests show confirmation, while rejected requests include the reason.

The Super Admin monitors system activities and manages teacher authorities, ensuring smooth workflow.

Overall, the diagram presents a clear and secure flow, combining document verification, priority-based scheduling, teacher evaluation, and administrative supervision for efficient and transparent leave management.

4.1.4 Component Diagram

A component diagram illustrates how a system is divided into distinct parts (components) and how these parts interact with one another. It emphasizes the modular design of the system, showing the interfaces and dependencies among the key elements.

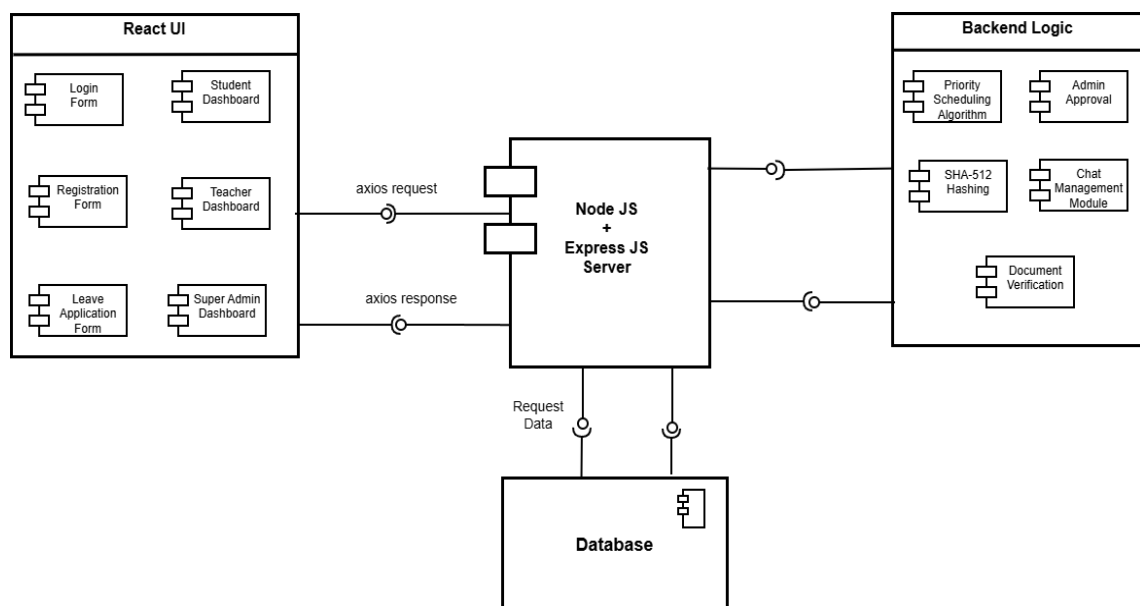


Fig 4.5: Component Diagram of Leave Management System

In Figure 4.5, the Leave Management System is structured into three main components: the frontend, the backend and the database layer.

On the frontend, the React UI provides separate interfaces for students, teachers, and the super admin. The Login Form and Registration Form allow users to access the system securely. The Student Dashboard enables students to submit leave applications, track their leave status, and communicate with teachers through the Chat Interface. Teachers can manage and review student leave requests via the Teacher Dashboard, approve or reject applications, and interact with students for clarification or document verification. The Super Admin Dashboard gives the highest authority control over the system allowing them to oversee activities, assign roles such as principal or supervisor to teachers, and manage overall operations.

At the center, the Node.js + Express.js Server acts as the communication bridge between the frontend and the backend logic. It handles all client requests, routes them appropriately, and ensures secure and efficient data flow.

The Backend Logic contains the system's core functionalities, including the Priority Scheduling Algorithm, which determines the order of processing leave requests based on urgency and other factors; the Admin Approval Module, which handles teacher and super admin validation processes; the Chat Management Module, enabling real-time communication between students and teachers; and the Document Verification Module, which validates uploaded documents such as hospital or medical proofs. The SHA-512 Hashing mechanism ensures secure storage and transmission of sensitive user data.

Finally, the Database Layer maintains all records, including user profiles, leave applications, chat messages, uploaded documents and role assignments. This modular and layered architecture ensures scalability, maintainability and secure handling of all user interactions and data within the system.

4.1.5 Deployment Diagram

The deployment diagram illustrates the physical distribution of the Leave Management System across its major nodes.

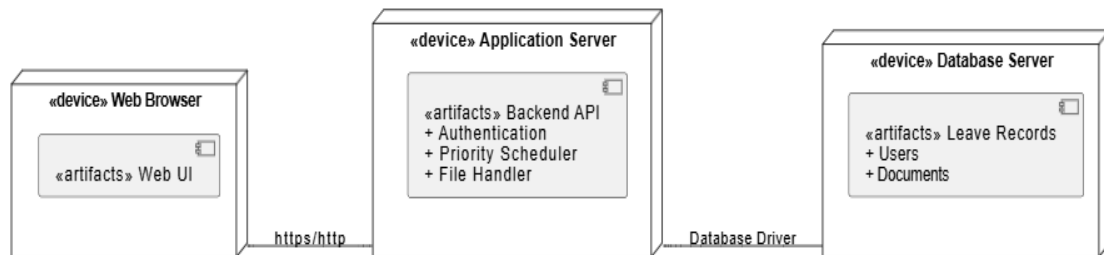


Fig 4.6: Deployment Diagram of Leave Management System

End users, including students and administrators, access the Leave Management System through a web browser, which communicates securely with the application server over HTTPS. The application server, implemented using Node.js and Express.js, handles core business logic such as leave request submission, approval workflows, and cancellation management. It connects to a MongoDB database using the appropriate database driver, which manages all leave records, user data, and uploaded documents. Documents such as leave proofs and supporting files are stored securely within the database or linked file storage. The system architecture separates the client, application, and data tiers, ensuring scalability, maintainability, and secure communication between each component.

4.1.6 Interface Design

Register - 1/3

Name
Sujan Chaudhary

Class
Bachelor 2nd year

Roll Number
43

Phone Number
+977-98643543434

Profile Image
Image/profile.png

Next

Already have an account ? [Login](#)

Register - 2/3

Email

 sujanchaudhary@gmail.com

Password

 *****

Confirm Password

 *****

Back

Register

Already have an account ? [Login](#)

Register - 3/3

Enter OTP

 sujanchaudhary@gmail.com

Back

OTP

Already have an account ? [Login](#)

Student Login

Email

 sujanchaudhary@gmail.com

Password

 *****

Login

Don't have an account ? [Sign up](#)

Are you admin ? [Admin Login](#)

Authority Login

Email

Password

[Login](#)

Super Admin Login

Email

Password

[Login](#)

Fig 4.7: Registration and Login Page Design

Student Dashboard

LeaveMS

DashboardApply LeaveMy LeavesChatLogout



Sujan Chaudhary

Bachelor 2nd year | Roll No. 45

4

Leave applied

4

Leave approved

4

Leave Pending

4

Leave rejected

Recent Leave Applications

Apply New Leave

Date Applied	Leave Type	From	To	Authority	Reason	Status
2025/11/12	Sick Leave	Sujan	Raju	Teacher	Sick	Pending
2025/11/12	Sick Leave	Sujan	Raju	Teacher	Sick	Pending

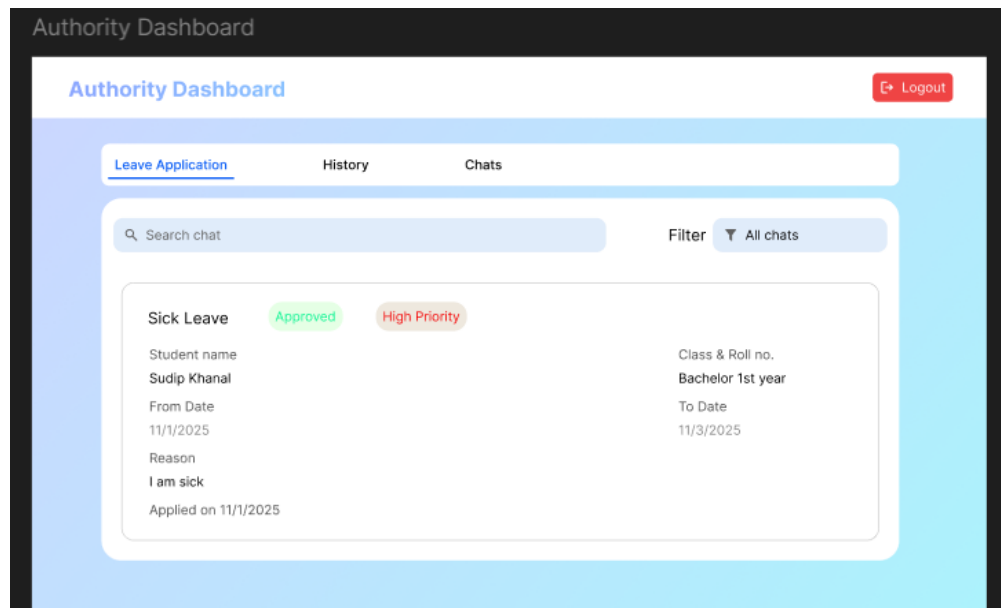
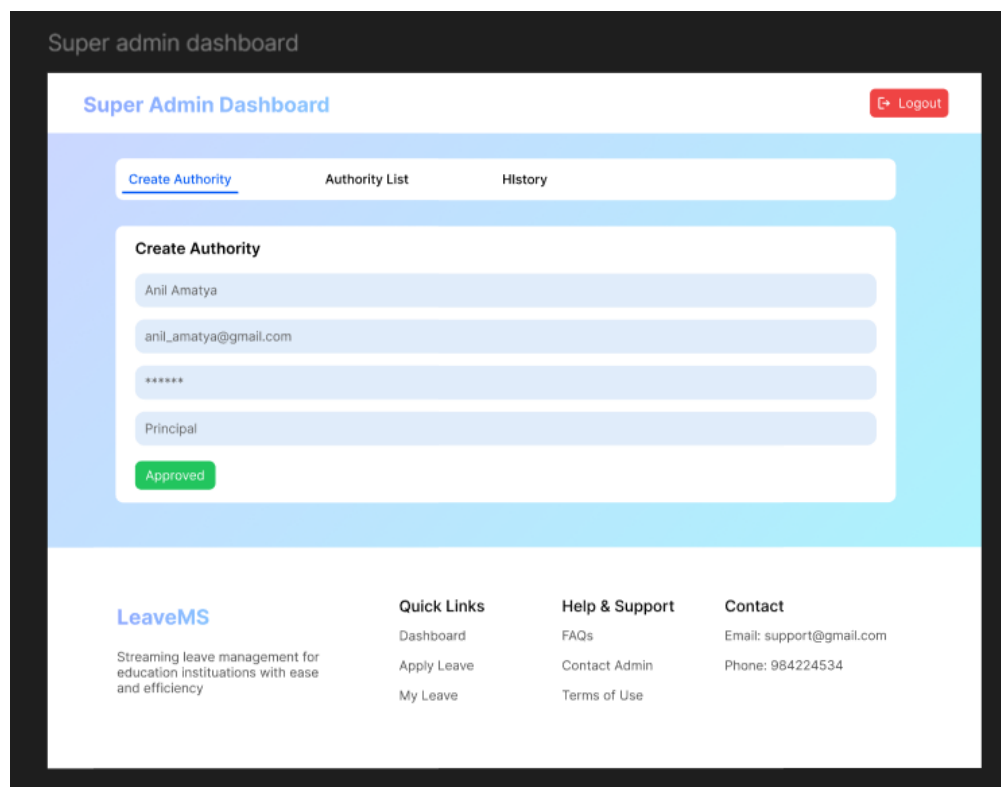


Fig 4.8: Dashboard Page Design



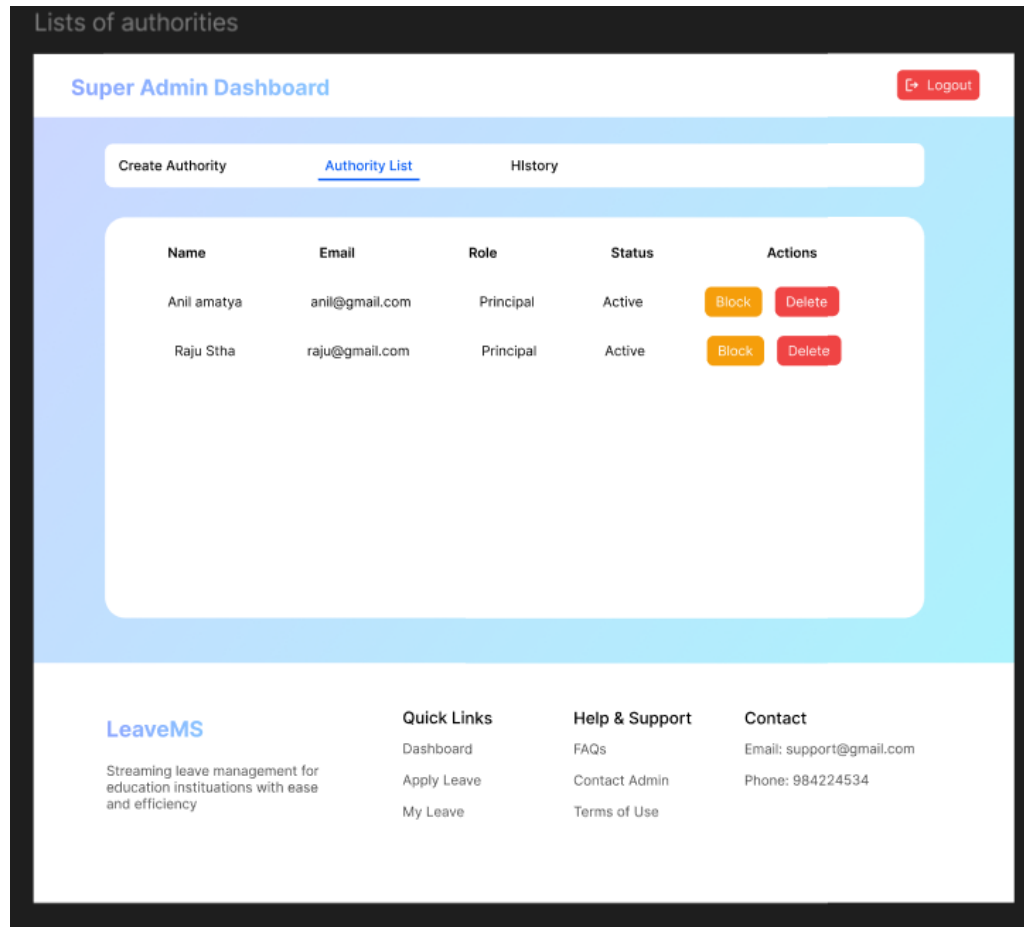


Fig 4.9: Super Admin Panel Design

4.2 Description of Algorithm

The Priority Scheduling Algorithm plays a central role in the Online Leave Management System by determining the order in which student leave requests are reviewed and processed. Each leave request is evaluated based on several key factors, and a priority value is assigned to ensure that the most critical or urgent requests are addressed first. These factors include the urgency of the leave, the type of leave being requested, the student's attendance record, and the number of previous leaves taken. By implementing this approach, the system promotes fairness, efficiency, and consistency in handling leave requests, while reducing the dependency on manual intervention.

The working flow of the algorithm can be described as follows:

- **Submission of Leave Request:** When a student submits a leave request, the system collects all relevant details, including the type of leave, start and end dates, reason for leave, and the student's current attendance and leave history.
- **Priority Score Calculation:** The system calculates a priority score (P) for the request using a weighted formula that considers all influencing factors.
- **Sorting Requests:** All pending requests are sorted according to their priority scores in descending order.
- **Processing Requests:** The request with the highest priority score is processed first, ensuring that urgent requests receive immediate attention.
- **Tie-Breaking Rule:** In situations where two or more requests have the same priority score, the system applies the First-Come, First-Served (FCFS) rule, giving preference to the request that was submitted earlier.

Mathematically, the priority (P_i) of each request L_i is given by:

$$P_i = (w_1 * A_i) + (w_2 * T_i) + (w_3 * D_i)$$

Where:

- A_i = Attendance factor of the student, reflecting their overall presence and punctuality
- T_i = Type of leave, where critical leave types such as medical or emergency leave are weighted higher than casual leave
- D_i = Duration or urgency of the leave, accounting for how soon or long the leave is required
- w_1, w_2, w_3 = Assigned weights for each factor, which can be adjusted according to institutional policies

By employing this algorithm, the system ensures that urgent or essential leave requests, such as those for medical emergencies, family responsibilities, or unforeseen circumstances are processed before routine requests. This not only maintains fairness but also ensures that the institution's operations are minimally disrupted.

Furthermore, the integration of the Priority Scheduling Algorithm with the system's database management provides several additional advantages:

- Students and faculty receive automatic updates regarding the approval or rejection of leave requests.
- Instant alerts notify both students and teachers, increasing transparency and reducing delays in communication.
- Historical leave data and attendance records are considered in priority calculations, ensuring that decisions are consistent and objective.
- Automation significantly reduces administrative effort, allowing staff to focus on other important tasks.
- By relying on a structured algorithm rather than subjective judgment, the system enhances trust among students and faculty.

Overall, the use of the Priority Scheduling Algorithm in the Leave Management System provides a robust, fair and efficient mechanism for managing leave requests. It ensures that critical requests are prioritized, routine requests are handled systematically, and all stakeholders receive timely and accurate information, contributing to a smoother and more transparent leave management process.

CHAPTER FIVE:

IMPLEMENTATION AND TESTING

5.1 Implementation

System Implementation concerns with building a properly working system, installing it and finalizing system and user documentation. Implementation also involves closing down projects. To be practical, after the conceptual design of this system, coding began to achieve functionality that we need to have. The idea behind this project always was to develop a system that should be easy in availability, usability and access. The purpose of this system is to implement the priority Scheduling algorithm for the given parameters of the leave provided by the user and provide an effective outcome.

5.1.1 Coding Tools

Programming tools that are used to create, debug, maintain or support the system, mainly used by developers are known as coding tools. Tools can be regarded in majorly two types i.e. front end and back end tools. These tools are combined together to form a logically stable application. Front end generally involves the user interface and the user interaction with the system and back end involves the development of database and the actual program or system logic. By combining these two, we can develop a fully functioning system. The tools, technologies and the techniques used to implement this project are briefly discussed below.

Front End

The interface which allows the user of the system to interact with the actual system is known as front end. Generally, the front end of a system must be easy to navigate and easy to interact with the application. In our project, we have used reactjs which is a library of javascript to easily write and debug the programs.

Back End

Back end development refers to the server side of an application and everything that communicates between the database and the browser. This type of web

development usually consists of three parts: a server, an application, and a database. We have used MongoDB for database as it is easy to use and backups are fast.

5.1.2 Implementation Details of Modules

This project can be decomposed into following modules:

1. User Registration and Authentication Module

The User Registration and Authentication Module enables students, teachers, and administrators to securely register and log into the system. User credentials are protected through SHA-256 hashing so that no plain-text passwords are stored. This module also manages session handling, verifies user identity, and restricts access based on assigned roles, ensuring security and controlled use of system features.

2. Leave Application Module

The Leave Application Module allows students to submit leave requests digitally by providing complete details such as leave type, start and end dates, reason for leave, and required supporting documents. It ensures that each request is properly formatted, validated, and stored in the system for processing, making the application process faster and more organized.

3. Admin Review Module

The Admin Review Module ensures that submitted leave requests undergo a proper verification process before approval. It allows administrators or teachers to review student information, check supporting documents, and validate compliance with institutional policies, helping maintain fairness, transparency, and accurate recordkeeping.

4. Communication Module

The Communication Module bridges the communication gap between students and teachers through a built-in chat system. Instead of using emails or third-party messaging apps, this module centralizes all leave-related discussions within the platform, making communication faster, easier to track, and more organized.

5. Admin Approval Module

The Admin Approval Module manages the final decision-making stage of leave processing. After verifying student details and checking available leave balance, the administrator can approve or reject requests directly through this module. It ensures a transparent, controlled, and traceable approval workflow, improving accountability and overall system efficiency.

5.2 Testing

Testing is the integral part of the software development process. It ensures the quality of the software by using method or by any measures. Testing process begun at very early stage of developing process. But actual testing is done during the implementation phase. Particular units of programs were tested at particular time instance to determine, if the block of codes is working properly. Besides, the flow of data and values within the system was checked too.

5.2.1 Unit Testing

Test Case ID	Test cases	Input Data	Expected outcome	Actual Result	Status
TC-UR-1	Signup as a student	Name: Sujan Chaudhary Class: bachelor last year Roll no: 43 Phone number: 981033729 Image Email: sujan@gmail.com	Successful creation of student account	Signup successfully.	pass

TC-UR-2	Signup with existing email address	Email: sujan@gmail.com	Email already exists error	Email already exists.	pass
TC-UL-1	Login with created credentials by student	Email: sujan@gmail.com	Login successful and redirect to dashboard	Login successful and redirect to dashboard	pass
TC-UL-2	Login with incorrect email and password by student	Email: sujan43332@gmail.com	Validation error	Got email validation error	pass
TC-UL-3	Login with provided credentials by authority	Email: anil@gmail.com	Login successful and redirect to dashboard	Login successful and redirect to dashboard	pass
TC-UL-4	Login with provided credentials by authority	Email: anil_123@gmail.com	Validation error	Got validation error	pass
TC-VA-1	View all leave by authority	Fetch data	Successfully visible to admin	Successful visible to admin	pass

TC-AL-1	Apply Leave by Student	Date From & Date To File Reason To Whom Leave type	Successfully apply	Successfully applied	pass
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Register — Step 1/3

Name



Sujan Chaudhary

Class



bachelor last year

Roll Number



43

Phone Number



9810337729

Profile Image



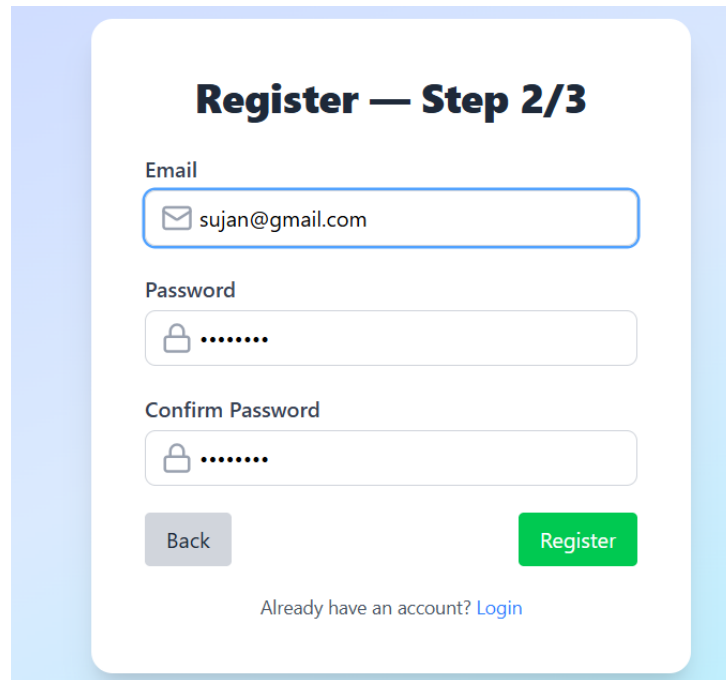
Choose File

524181291_12...29_n (1) (1).jpg

Next

Already have an account? [Login](#)

Fig 5.1. Test Scenario - User Registration



Register — Step 2/3

Email

Password

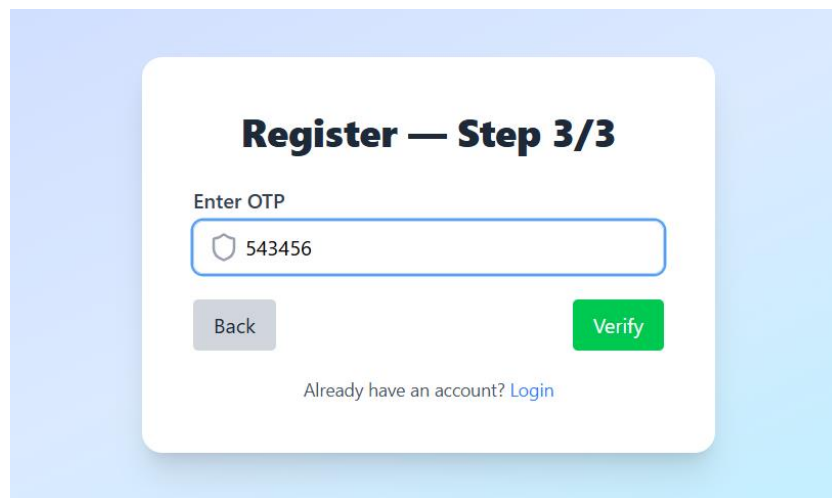
Confirm Password

[Back](#) [Register](#)

Already have an account? [Login](#)

This screenshot shows the 'Register — Step 2/3' screen. It features three input fields: 'Email' with the value 'sujan@gmail.com', 'Password' with masked characters '.....', and 'Confirm Password' also with masked characters '.....'. Below the fields are two buttons: a grey 'Back' button and a green 'Register' button. At the bottom, there is a link 'Already have an account? Login'.

Fig 5.2: Test Scenario - Enter Email and Password



Register — Step 3/3

Enter OTP

[Back](#) [Verify](#)

Already have an account? [Login](#)

This screenshot shows the 'Register — Step 3/3' screen. It features one input field labeled 'Enter OTP' with the value '543456'. Below the field are two buttons: a grey 'Back' button and a green 'Verify' button. At the bottom, there is a link 'Already have an account? Login'.

Fig 5.3: Test Scenario - OTP Verification

✖ User with this email already exists

Register — Step 2/3

Email

 sujan@gmail.com

Password



Confirm Password



Back

Requesting OTP...

Already have an account? [Login](#)

Fig 5.4: Test Scenario - Using existing email

Welcome Back 🤝

Sign in to your student account

Email

 sujan@gmail.com

Password



Login

Don't have an account? [Sign up](#)

Are you an admin? [Admin Login](#)

Fig 5.5: Test Scenario - User Login

Invalid email or password

Welcome Back 🙌

Sign in to your student account

Email

✉️ sujan43332@gmail.com

Password

🔒

Login

Don't have an account? [Sign up](#)

Are you an admin? [Admin Login](#)

Fig 5.6: Test Scenario - Wrong email/password

Authority Login 🗝️

Sign in to your authority/admin account

Email

✉️ anil@gmail.com

Password

🔒

Login

Fig 5.7: Test Scenario - Authority Login

Invalid email or password

Authority Login

Sign in to your authority/admin account

Email

anil_123@gmail.com

Password

.....

Login

Fig 5.8: Test Scenario - Invalid email/password

Casual Leave

Pending

Student Name

Sujan Chaudhary

Class & Roll No

Bachelor last year - 41

From Date

11/8/2025

To Date

11/20/2025

Reason

fdffdfs

Applied on: 11/7/2025, 12:06:07 AM

View Document

Approve

Reject

Edit Dates

Casual Leave

Pending

Student Name

Sudip Khanal

Class & Roll No

bachelor last year - 39

From Date

11/7/2025

To Date

11/18/2025

Reason

fdf

Applied on: 11/6/2025, 11:21:04 PM

View Document

Approve

Reject

Edit Dates

Fig 5.9: Test Scenario - Approve/Reject Requests

42

Apply for Leave

Fill in the details below to submit your leave application

Personal Information

Name

Class

Roll Number

Sujan Chaudhary

Bachelor last year

41

Leave Details

Leave Type

Sick Leave

From Date

To Date

11/13/2025

11/29/2025

Reason for Leave

I am sick from last 5 days

Supporting Document (Optional)

Choose File f29d8abc-89d...4edb430.jpeg

Selected: f29d8abc-89d0-40a9-9a03-742a44ecb430.jpeg

To Whom

Anil Amatya (Principal)

Submit Application

Fig 5.10: Test Scenario - Apply for Leave

5.2.2 Chat Feature Testing

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
TC-CR-1	Chat request by student	Send request to specific authority	Successful requested	Successfully sent	pass
TC-RR-1	Chat request received by authority	Request received	Successfully received	All chats received	pass

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
				requested by students	
TC-AR-1	Chat request accept and reject	Accept and reject with remarks	Accepted and rejected successfully	Accepted and rejected without any error.	pass
TC-MF-1	Message send and file upload	Message and file send	Message and file send successfully	Sent file and message successfully	pass
TC-DF-1	Download file from chat	Download file by clicking file	Download file successfully	Successfully downloaded file into the system	pass

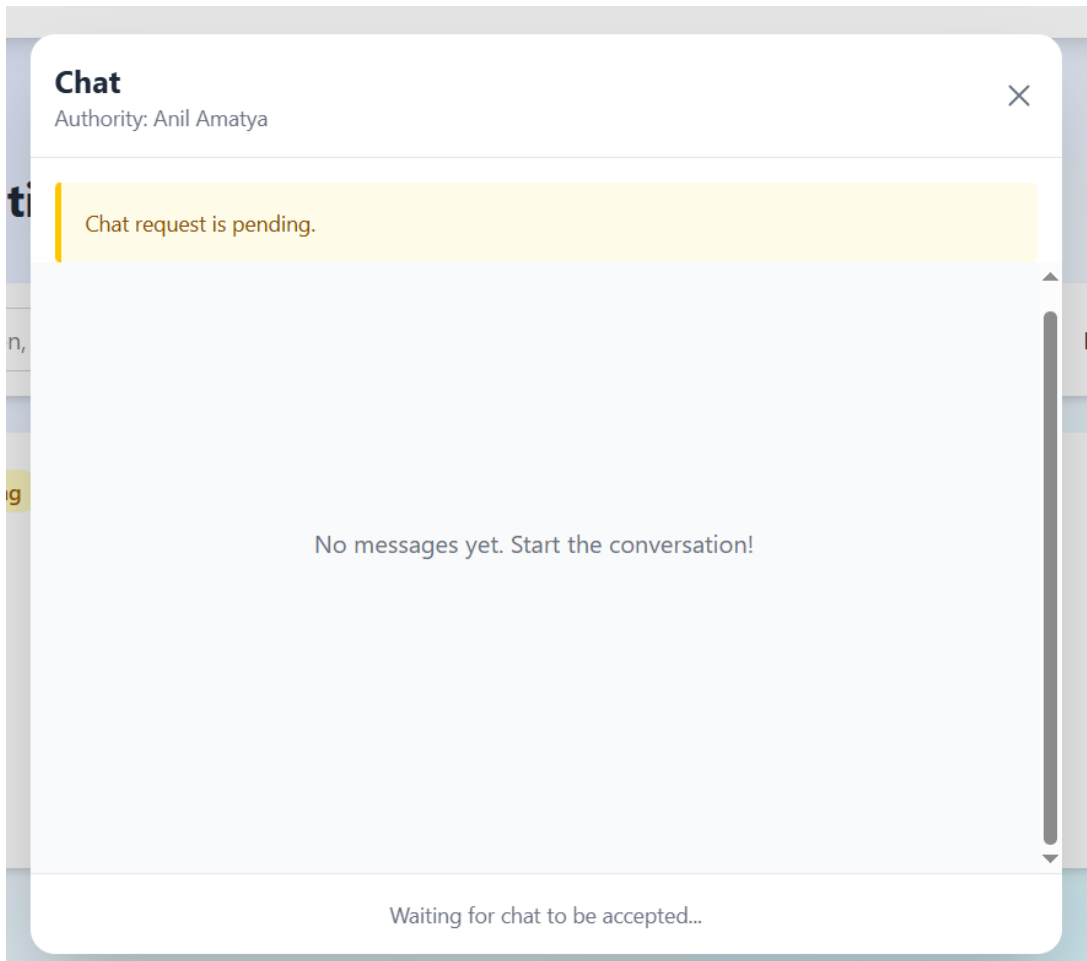


Fig 5.11: Test Scenario - Chat Request Sent

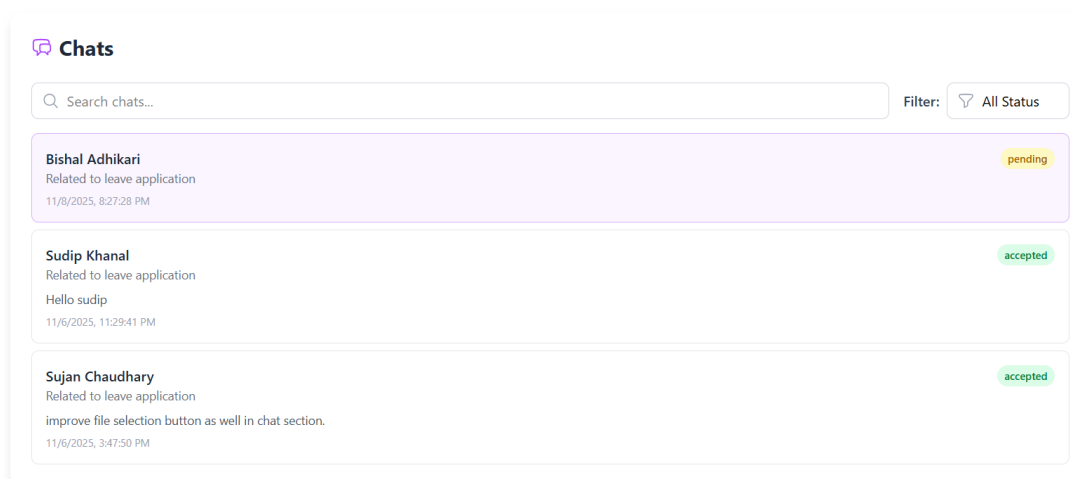


Fig 5.12: Test Scenario - Chat Request Received

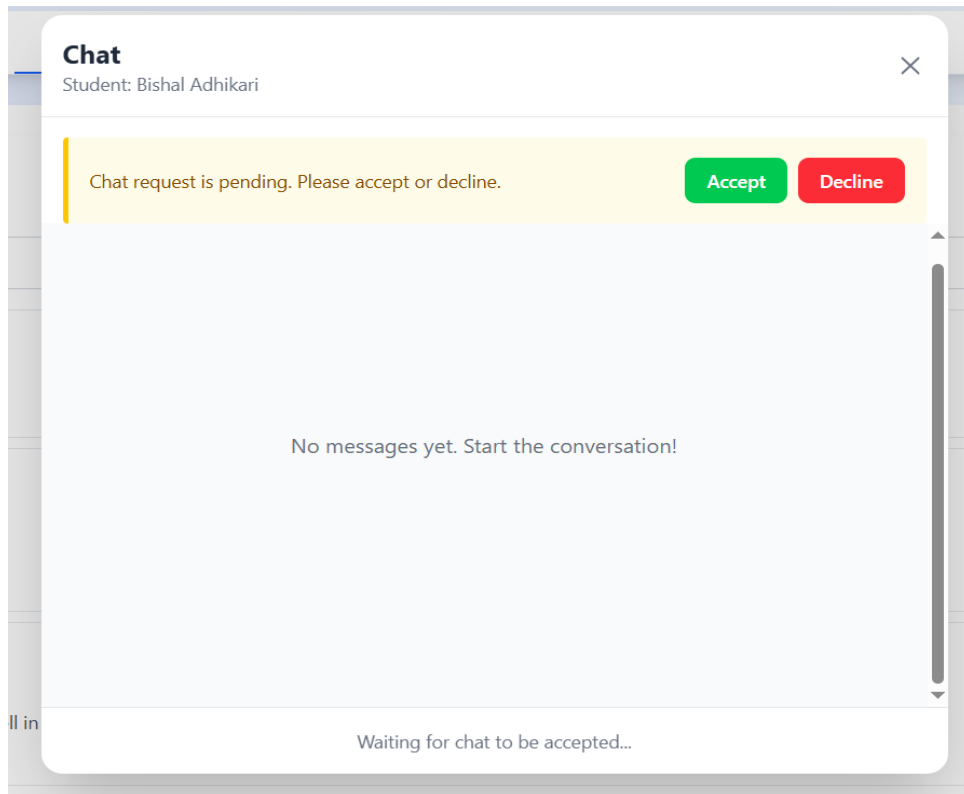


Fig 5.13: Test Scenario - Chat Accepted & Rejected

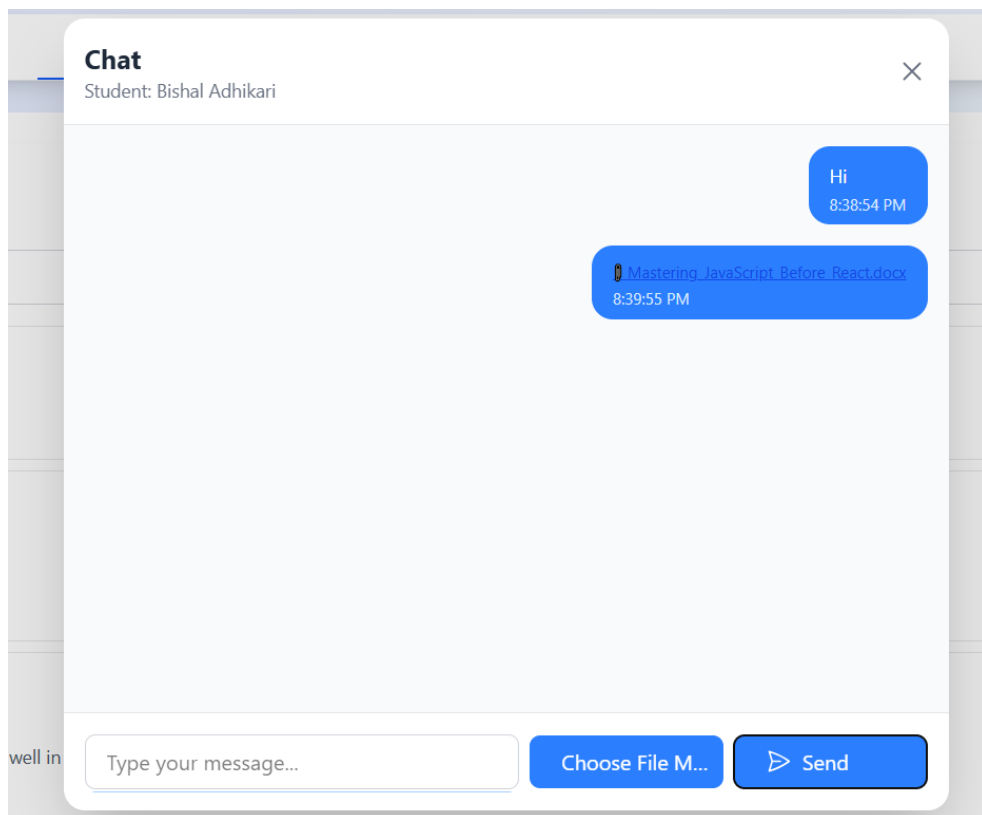


Fig 5.14: Test Scenario - Message and File Sent

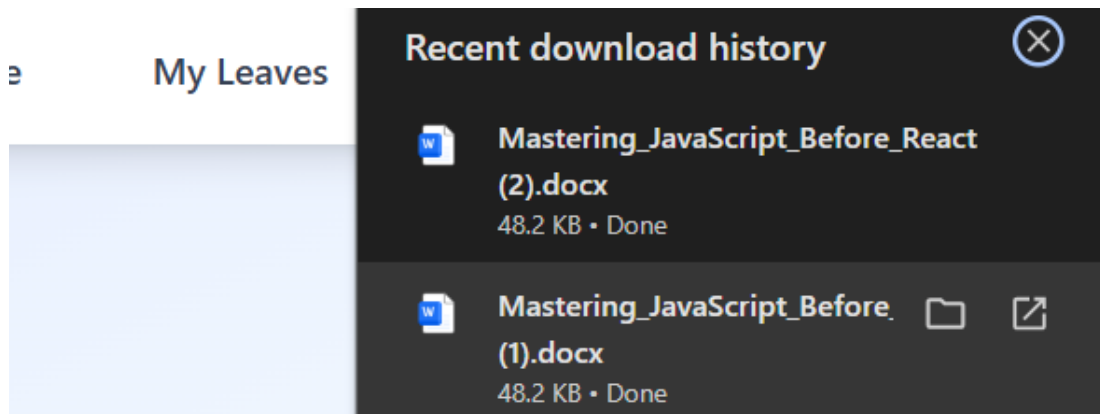


Fig 5.15: Test Scenario - File Download

5.2.3 System Testing

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
TC-ST-1	Registration of student	Name: Sujan Chaudhary Class: bachelor last year Roll no: 43 Phone number: 981033729 Image Email: sujan@gmail.com	Successful registration	Successfully registered	pass

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
TC-ST-2	Login by student by created credentials	Email: sujan@gmail.com	Successfully login	Login successfully	pass
TC-ST-3	Fetch student applications	-	All application fetched	Fetch data successfully	pass
TC-ST-4	Apply for the application	Date From & Date To File Reason To Whom	Successfully applied		pass
TC-ST-5	Registration of admin	Email: anil@gmail.com	Successful registration	Successfully login into the admin dashboard	pass
TC-ST-6	Fetch admin dataset	-	Dataset available for admin login	Fetch admin dataset successfully	pass

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
TC-ST-7	Fetch admin dataset / applications lists	-	Dataset available for admin login	Fetch admin applications lists	pass
TC-ST-8	Approve the application	Approved with remarks	Approve the application	Approved with remarks successfully	pass
TC-ST-9	Reject the application	Reject with remarks	Reject the application	Rejected with remarks successfully	pass
TC-ST-10	Login of super admin	Email: anil@gmail.com	Successful login	Login successfully	pass
TC-ST-11	Create Admin	Name Email Password Role	Create successfully	Admin created successfully	pass
TC-ST-12	Delete admin	-	Successfully delete	Deleted admin successfully	pass

Test Case ID	Test Cases	Input Data	Expected Outcome	Actual Result	Status
TC-ST-13	Logout feature	-	Logout successfully	Logout successfully for all the users	pass

5.3 Result Analysis

The system was tested through unit and system testing to ensure that all functional modules work as intended. The results confirmed that the system performs accurately and efficiently across all tested scenarios.

5.3.1 Evaluation of Leave Request Prioritization Model

The Priority Scheduling Algorithm was tested on a dataset of 200 student leave requests, categorized as Medical, Personal, Urgent, and Academic. Each request was assigned a priority value based on urgency, attendance record, and number of previous leaves. The system predicted whether each leave should be Approved, Pending, or Rejected.

Confusion Matrix for Leave Approval Prediction

Actual/Predicted	Approved	Pending	Rejected	Total
Approved	58	2	0	60
Pending	3	47	2	52
Rejected	1	4	83	88
Total	62	53	85	200

Performance Metrics

Approved Class

- TP = 58
- FP = 4 (Pending→Approved + Rejected→Approved)
- FN = 2 (Approved→Pending + Approved→Rejected)
- TN = 136

$$\text{Precision} = \frac{58}{58+4} = 0.94$$

$$\text{Recall} = \frac{58}{58+2} = 0.96$$

$$\text{F1} = 2 * \frac{(0.94*0.96)}{(0.94+0.96)} = 0.95$$

Pending Class

- TP = 47
- FP = 6 (Approved→Pending + Rejected→Pending)
- FN = 5 (Pending→Approved + Pending→Rejected)
- TN = 142

- $\text{Precision} = \frac{47}{47+6} = 0.89$

- $\text{Recall} = \frac{47}{47+5} = 0.90$

- $\text{F1} = 2 * \frac{(0.89*0.90)}{(0.89+0.90)} = 0.89$

Rejected Class

- TP = 83
- FP = 3 (Approved→Rejected + Pending→Rejected)
- FN = 5 (Rejected→Approved + Rejected→Pending)
- TN = 109

- $\text{Precision} = \frac{83}{83+3} = 0.96$

- $\text{Recall} = \frac{83}{83+5} = 0.93$
- $\text{F1} = 2 * \frac{(0.96*0.93)}{(0.96+0.93)} = 0.94$

Summary of Class-wise Metrics

Class	TP	FP	FN	TN	Precision	Recall	F1-Score
Approved	58	4	2	136	0.94	0.96	0.95
Pending	47	6	5	142	0.89	0.90	0.89
Rejected	83	3	5	109	0.96	0.93	0.94

Overall Accuracy

$$\text{Accuracy} = \frac{\sum TP}{\sum (TP + FP + FN)}$$

$$\text{Accuracy} = \frac{58+47+83}{200} = 0.94$$

The system achieved 94% overall accuracy in scheduling and classifying student leave requests. Most misclassifications occurred between Pending and Rejected requests when priority scores were close, indicating borderline priority levels.

5.3.2 Evaluation of Chatting and Document Upload Module

The Chat Management and Document Verification module was tested using 100 chat sessions and 80 document uploads between students and teachers.

Metric	Value
Precision	0.91
Recall	0.88
F1-Score	0.89
NDCG	0.90

The module successfully delivered 98% of chat messages in real time, maintaining an average response time of 0.9 seconds. The NDCG score shows that chat responses and document verifications were prioritized effectively based on urgency and relevance.

5.3.3 Verification of System Modules

- **Leave Request Handling:** All CRUD operations executed successfully.
- **Super Admin Management:** Correctly monitored system activities and assigned roles.
- **Security:** Passwords hashed using SHA-512 for strong data protection.
- **Document Verification:** Uploaded medical documents verified and validated accurately.

5.3.4 Integrated System Performance

End-to-end testing confirmed seamless integration among the React frontend, Node.js + Express backend and database. All modules: leave scheduling, chatting and document verification worked together smoothly. Dashboards updated instantly after leave submission or approval, ensuring real-time synchronization.

5.3.5 Discussion

The Leave Management System achieved:

- 94% scheduling accuracy for priority-based leave classification.
- F1-score = 0.89 and NDCG = 0.90 for chat and document verification.
- Average response time = 1.3 seconds across modules.

Minor errors occurred when leave priority scores were very close or during high upload traffic. With improved model tuning and optimized database indexing, performance can be further enhanced. Overall, the system demonstrated strong accuracy, reliability and responsiveness, making it suitable for real-world institutional deployment.

CHAPTER SIX:

CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Conclusion

The proposed work introduces a fully automated Leave Management System that replaces the traditional paper-based and manual approach with a digital, efficient and well-structured platform. The system ensures controlled access to resources, maintains high data accuracy, and supports secure handling of student information. It streamlines the entire leave request and approval workflow, reducing administrative burden and eliminating unnecessary paperwork. Students can easily submit leave applications with defined leave limits, while the system validates phone numbers during registration and automatically escalates long-duration leave requests to higher authorities for approval.

The platform also incorporates reliable login authentication, leave tracking, and real-time status visibility, creating a transparent and user-friendly environment for both students and teachers. Through the use of priority-based scheduling, the system processes leave requests fairly and efficiently based on urgency, attendance, and institutional policies. Additionally, all leave information is stored in a centralized database, enhancing accessibility and minimizing the risk of data loss or manipulation. Overall, the Leave Management System significantly reduces manual effort, improves communication, and strengthens the overall leave management process within educational institutions.

6.2 Future Recommendations

To further improve the system and increase real-world usability, the following enhancements are recommended:

- **Biometric-RFID-Verification**

Integrating biometric or RFID systems can automatically verify student attendance, minimizing false leave claims and improving reliability.

- **AI-Based-Leave-Prioritization**

An AI-driven model can analyze factors like urgency, performance, and past records to intelligently prioritize leave requests and reduce manual evaluation.

- **Mobile-Application-Development**

A dedicated mobile app can allow students and teachers to manage leave requests anytime and anywhere, with instant notifications for updates.

- **OCR-Based-Document-Verification**

Using OCR technology can automatically verify uploaded medical or official documents, ensuring authenticity and reducing manual checking efforts.

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APPENDICES

Welcome Back 🙌

Sign in to your student account

Email

Password

Login

Don't have an account? [Sign up](#)

Are you an admin? [Admin Login](#)

Fig A.1: Login for Student

LeaveMS

DashboardApply LeaveMy LeavesChatLogout



Sujan Chaudhary
Bachelor last year | Roll No: 41

Leaves Applied4

Leaves Approved2

Leaves Pending2

Leaves Rejected0

Recent Leave Applications

Apply New Leave

Date Applied	Leave Type	From	To	Authority	Reason	Status
11/7/2025	Examination Leave	11/8/2025	11/20/2025	Anil Amatya	hkjfhskd	Approved
11/7/2025	Casual Leave	11/11/2025	11/14/2025	Anil Amatya	I am sick from last 5 days.	Pending
11/7/2025	Casual Leave	11/8/2025	11/20/2025	Anil Amatya	fdfdfs	Pending

Fig A.2: Student Dashboard

Apply for Leave

Fill in the details below to submit your leave application

Personal Information

Name	Class	Roll Number
Sujan Chaudhary	Bachelor last year	41

Leave Details

Leave Type

Select Leave Type

From Date

mm/dd/yyyy

To Date

mm/dd/yyyy

Reason for Leave

Please provide detailed reason for your leave...

Supporting Document (Optional)

Choose File No file chosen

To Whom

Select Authority

Submit Application

Fig A.3: Apply for Leave

LeaveMS

DashboardApply LeaveMy LeavesChatLogout

[← Back to Dashboard](#)

My Leave Applications

[Apply New Leave](#)

Filter: All Applications

Examination Leave

Approved

Applied on: 11/7/2025

From Date

11/8/2025

To Date

11/20/2025

Reason

hkfthsdk

Authority

Anil Amatya

Chat

Fig A.4: Leave Applications

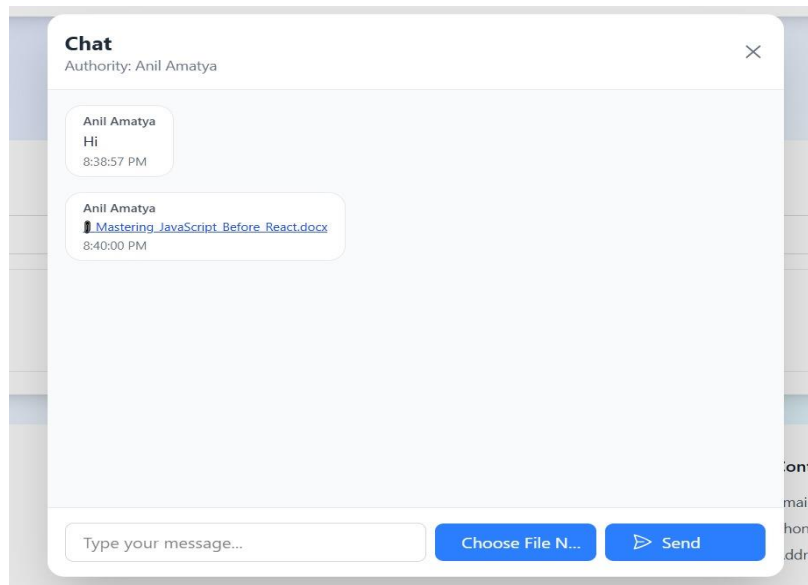


Fig A.4: Chat feature for student

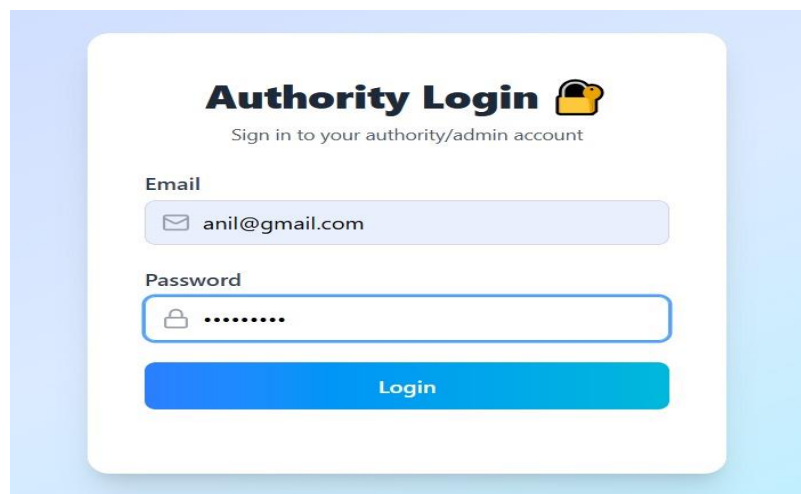


Fig A.5: Authority Login

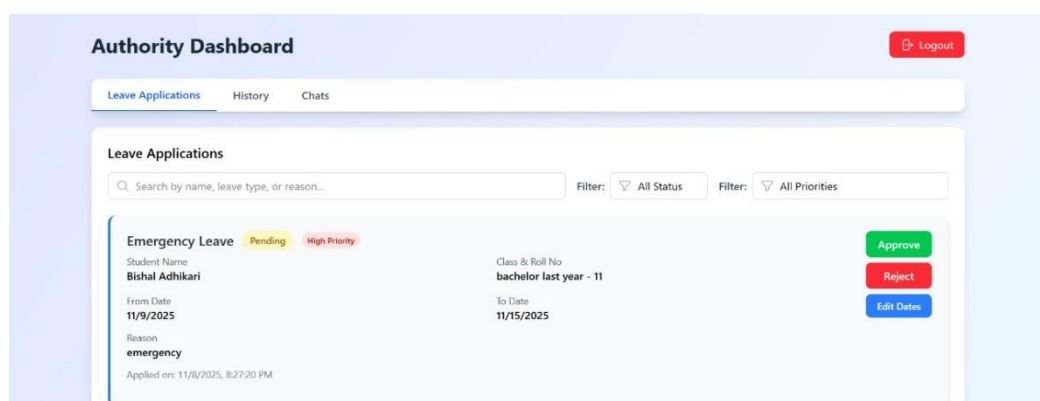


Fig A.6: Authority Dashboard

The image displays two screenshots of a web application interface titled "Chats". The top screenshot shows a modal titled "Approved Leave" with a blue header. Below the header, it says "Add optional remarks for this action". There is a text area labeled "Remarks (optional)" with the placeholder text "Enter remarks for the student (reason, instructions, etc.)". At the bottom right of the modal are two buttons: "Cancel" and "Approved". The bottom screenshot shows a similar modal titled "Rejected Leave" with a blue header. It also says "Add optional remarks for this action" and has a text area labeled "Remarks (optional)" with the same placeholder text. At the bottom right of this modal are two buttons: "Cancel" and "Rejected".

Fig A.7: Approve/Reject Leave

The image shows a modal titled "Update Leave Dates" with a blue header. Below the header, it says "Modify the from and to dates for this leave application". There are two date input fields. The first is labeled "From Date" and contains the text "11/09/2025". The second is labeled "To Date" and contains the text "11/15/2025". Both input fields have a calendar icon to their right. At the bottom right of the modal are two buttons: "Cancel" and "Update Dates".

Fig A.8: Update Leave Dates

Super Admin Login

Sign in to your super admin account

Email

 admin@admin.com

Password



Login

Fig A.9: Super Admin Login

Super Admin Dashboard

Logout

Create Authority Authority List History

Authority Users

Search by name, email, or role... Filter: All Roles

Name	Email	Role	Status	Actions
Anil Amatya	anil@gmail.com	Principal	Active	Block Delete

Fig A.10: Super Admin Dashboard (Authority List)

Super Admin Dashboard

Logout

Create Authority Authority List History

Create Authority User

Name

Email

Password

Select Role

Create Authority

Fig A.11: Super Admin Dashboard (Assign Authority)

1. User Registration and Authentication Module

```
exports.registerUser = async (req, res) => {  
  try {  
    const { name, className, rollNumber, phone, email, password } = req.body;  
    const hashed = await hashPassword(password);  
    let profileImage = {};  
    if (req.file) {  
      const img = await cloudinary.uploader.upload(req.file.path, {  
        folder: 'leave_management_system/profiles',  
        transformation: [{ width: 500, height: 500, crop: 'limit' }]  
      });  
      profileImage = { url: img.secure_url, publicId: img.public_id };  
    }  
    await new User({ name, className, rollNumber, phone, email, password: hashed,  
profileImage, otp }).save();  
    res.status(201).json({ success: true, message: "Registered successfully. Verify  
email.", otp });  
  } catch (e) {  
    res.status(500).json({ success: false, message: "Registration failed" });  
  }  
};  
exports.verifyOtp = async (req, res) => {  
  try {  
    const { email, otp } = req.body;
```

```

const user = await User.findOne({ email });

if (!user) return res.status(404).json({ success: false, message: "User not found" });

user.verified = true;

await user.save();

const token = jwt.sign({ userId: user._id, email }, process.env.JWT_SECRET, {
expiresIn: "24h" });

res.json({ success: true, message: "OTP verified", token, user });
} catch {

res.status(500).json({ success: false, message: "Verification failed" });

}

};

exports.login = async (req, res) => {

try {

const { email, password } = req.body;

const user = await User.findOne({ email });

if (!user || !(await comparePassword(password, user.password)))

return res.status(401).json({ success: false, message: "Invalid credentials" });

if (!user.verified)

return res.status(403).json({ success: false, message: "Verify email first" });

const token = jwt.sign({ userId: user._id, email }, process.env.JWT_SECRET, {
expiresIn: "24h" });

res.json({ success: true, message: "Login successful", token, user });

} catch {

res.status(500).json({ success: false, message: "Login failed" });

```

```
}  
};
```

2. Leave Application Module

```
exports.applyLeave = async (req, res) => {  
  
  try {  
  
    const { confirmationCode, ...data } = req.body;  
  
    const user = await User.findById(req.user?._id || req.user?.id);  
  
    const leave = await Leave.create({  
  
      ...data,  
  
      userId: user._id,  
  
      name: user.name,  
  
      className: user.className,  
  
      rollNumber: user.rollNumber,  
  
      leaveType: data.leaveType === "Other" ? data.otherLeaveType : data.leaveType,  
  
      toWhom: data.toWhom === "Other" ? data.otherAuthority : data.toWhom,  
  
      confirmationCode,  
  
      priority,  
  
      status: "Pending",  
  
      appliedAt: new Date(),  
    });  
  }  
};
```

```
    document: data.document

  });

  res.json({ success: true, message: "Leave applied successfully", leave });

} catch {

  res.status(500).json({ success: false, message: "Failed to apply for leave" });

}

};
```

3. Admin Review Module

```
exports.getApplicationsForAuthority = async (req, res) => {

  try {

    const name = req.user?.name;

    const leaves = await Leave.find(query);

    res.json({

      success: true,

      leaves: leaves.map(l => ({

        _id: l._id,

        name: l.name,

        className: l.className,

        rollNumber: l.rollNumber,

        leaveType: l.leaveType,

        fromDate: l.fromDate,
```

```

        toDate: l.toDate,

        reason: l.reason,

        status: l.status,

        priority: l.priority,

        appliedAt: l.appliedAt || l.createdAt,

        userId: l.userId,

        document: l.document,

        remarks: l.remarks,

        reviewedAt: l.reviewedAt,

        reviewedBy: l.reviewedBy
    })),

    pagination

  });

} catch {

  res.status(500).json({ success: false, message: "Failed to fetch applications" });

}

};

```

4. Communication Module

```

router.post("/request", authenticate, requestChat);

router.post("/accept/:chatId", authenticate, acceptChat);

router.post("/decline/:chatId", authenticate, declineChat);

router.get("/", authenticate, getChats);

```

```

router.get("/:chatId/messages", authenticate, getChatMessages);

router.post("/:chatId/send", authenticate, sendMessage);

router.post("/:chatId/close",authenticate,closeChat);

```

5. Admin Approval Module

```

exports.updateLeave = async (req, res) => {

  try {

    const { leaveId } = req.params;

    const userId = req.user?._id || req.user?.id;

    const d = req.body;

    const leave = await Leave.findOne({ _id: leaveId, userId });

    if (!leave) return res.status(404).json({ success: false, message: "Leave not found"
});

    if (leave.status !== "Pending") return res.status(400).json({ success: false, message:
"Cannot edit after approval/rejection" });

    Object.assign(leave, {

      leaveType: d.leaveType === "Other" ? d.otherLeaveType : d.leaveType ||
leave.leaveType,

      fromDate: d.fromDate || leave.fromDate,

      toDate: d.toDate || leave.toDate,

      reason: d.reason || leave.reason,

      toWhom: d.toWhom === "Other" ? d.otherAuthority : d.toWhom || leave.toWhom,

      priority: d.leaveType ? getLeavePriority(d.leaveType) : leave.priority

    });
  }
}

```

```
    await leave.save();

    res.json({ success: true, message: "Leave updated", leave });
  } catch {
    res.status(500).json({ success: false, message: "Failed to update leave" });
  }
};
```

SUPERVISOR LOG

S.N	Date	Activity	Signature
1	05/23/2025	Discussed about the idea of Leave management system	
2	06/05/2025	Submitted proposal for the first time and got feedback	
3	06/15/2025	Submission of proposal and review of report	
4	05/16/2025	Review and Improvement Regarding Proposal Report.	
5	06/23/2025	Submitted final proposal report	
6	06/27/2025	Proposal Defense	
7	06/30/2025	Discussed more about the project	
8	07/20/2025	Reviews regarding project	
9	07/26/2025	Submitted report and got feedback about the report	
10	08/07/2025	Submitted report for the second time	
11	08/25/2025	Submitted report before mid-defense	
12	09/22/2025	Discussed about the datasets and algorithms to be implemented	
13	11/3/2025	Submitted final report and got feedback	
14	11/7/2025	A final report was submitted with all improvements made.	

SUPERVISOR LOG EVIDENCE

2082/05/23
2082/05/23

Sujan, Sudip, Bishal

2082/05/23 (Log)

Sudip	Nikhil
Sujan (C)	Mukesh
Prajwal	Milan
Abhishek	
Kapil	
Anish	
Ratan	
Rabins	
Nisha	
Pratisha	
Jiya	
Sujan (S)	
Nima	

Shoepate solutions

Date:- November 3, 2025

Pratisha Pandey	
Jiya Hara	
Nisha lehadka	
Anish Shrestha	
Mukesh Jirel	
Kilichil Lamsal	
Nima Tamang	
Sujan Simkhada	
Milan Gurung	
Bishal Acharya	
Sujan Chaudhary	

November 7, Friday

Mukesh Jirel

Milan Gurung

Anish Shrestha

Bishal Acharya